TFPT — Frontier items

η_B, m_p/m_e, the Koide relation, dark matter and quantum gravity — the honest status, not forced onto the ladder

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October 18, 2026 · v5.4 (rev 8)

What this document is about — the status authority for the frontier

The honest frontier: which physics has a genuine TFPT handle and which does not. For each of η_B, m_p/m_e, the Koide relation, dark matter and full quantum gravity, this note states the genuine structural handle, the precision with which it currently lands, and — crucially — what is *not* a clean compiler power and is deliberately *not* forced onto the ladder. **This document is the status authority for the frontier items:** where any other document's wording and the markers here disagree, the typing here (and the `status_ledger.csv`) wins.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and four companions** (plus one standalone working note), best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

Working notes: N1. `note_e8_gaussian_code` — the Gaussian code bridge: E_8 over $\mathbb{Z}[i]$ via Construction A over the extended Hamming code $[8, 4, 4]$, the canonical four-bit quotient $L/(1+i)L \cong \mathbb{F}_2^4$, and the G_{31} quartic companion (exact algebra, machine-certified).

You are reading document 4 (Frontier).

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); CLOSED METROLOGY ($\mathbb{R}_+$ torsor; one dim. calibration by dimensional analysis; No-Unit v153/v725)	closed metrology [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$	[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

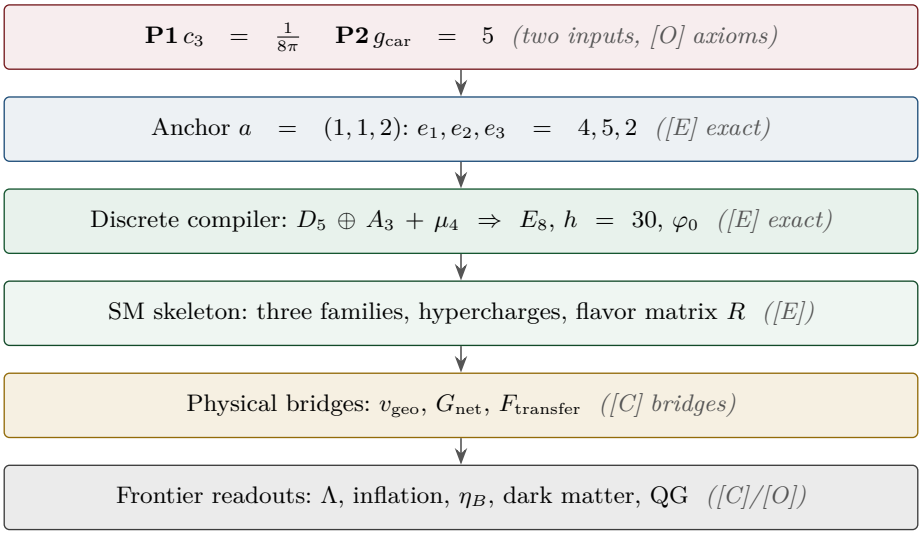
Beside that compiler rest sits the strict-physical-TOE rest: the ten-component Rest_{TOE} / the eight gates of `TFPT.TOE.COMPLETE.01` [O] ($\text{TOE}_{\text{Complete}} :=$ AND(T1..T8); not a closure).

The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. The classical metric-sector field equation is no longer only an $R{+}R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2|2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2|\chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status (an external candidate action — Bianconi's entropic action, arXiv:2408.14391 — is quantified in `v473` without changing the typing) and the one unit v_{geo}; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem G_{net} (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) now carries two named routes (route split 2026-07-22, no status change in substance): the MMST route `SEAM.EQUIV.MMST.01` is *closed modulo cited theorems*; the twistor route `SEAM.EQUIV.TWISTOR.01` (Costello–Li, prepared by `CELEST.SEAM.01`) and the unconditional parent stay [O]. On the MMST route: the explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD $= 1$ (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual of that route that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package ($h_s{=}16/16{=}1{\in}\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM $\mu{=}4/2^2{=}1$; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed'` (`v469`); so the MMST route is *closed modulo cited theorems*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76`/`v330`). So the honest residual is the one unit v_{geo} plus the typed F_{transfer} interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus F_{\text{transfer}}$, with the seam G_{net} closed modulo cited theorems on its MMST route (parent [O]) and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test



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Ground rule. These five are exactly the items previously flagged "*not on a clean action/ladder yet — do not force.*" This note gives, for each, the *genuine* structural handle that does exist, the precision with which it currently lands, and a clear statement of what is *not* a single compiler power. No fabricated ladder rungs; the honest markers [E]/[C]/[O] are applied throughout. (All numbers machine-verified.)

Hard rule on the frontier — one interface F_{transfer}

Koide, η_B, the axion relic scale and m_p/m_e are *not* compiler powers unless their missing QFT-transfer / cosmology computation is explicitly supplied. They are deliberately typed interfaces, not structural holes, and they are *four instances of one missing functor* F_{transfer} — the continuous transport from compiler *source* data to measured observables (Koide source→pole; η_B source→Boltzmann relic; axion scale→cosmological abundance; $m_p/m_e \to$ QCD/EW matching). Each has a genuine handle (a near-value, a determined readout, a fixed candidate) but its *exact* status is conditional on that named transfer calculation — now a runnable [C] solver with a kill test (v371–v375), still a transfer target, never a compiler power. In particular η_B is a *transfer target*, not a primitive compiler prediction. They must never be quoted as primitive TFPT outputs.

Why this is principled, not a gap: these are precisely the **discrete↔ continuous boundary** of the theory — the points where the discrete compiler hands its skeleton off to continuous physics (QCD confinement, cosmological Boltzmann transport, relic integrals). The cleanness of a quantity tracks how

far it sits from that handoff: pure compiler ratios are exact, QCD/cosmology-dressed quantities are typed interfaces. [verification/v35_quark_qcd_boundary.py]

Machine-enforced. This rule is a CI-style guard ([verification/v187_ftransfer_laws.py]): the four frontier families (F_{pole} Koide, $F_{\text{Boltzmann}}$ η_B, F_{relic} axion, F_{QCD} m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — exact algebraic sub-parts (the Koide 53/54 factor, $b_3 = -7$) may be [E], but the *physical* prediction never is. No future edit can quietly promote a frontier number to a closed compiler output. A companion *prose* sentinel ([verification/v188_frontier_wording_guard.py]) forbids stale sentences whose semantics an earlier round superseded, so a frozen release can never re-introduce wording that was true yesterday and is half-true today.

The functor contract `CONTRACT.F.01`. Beyond the typing guard, the functor is pinned by four checkable structural axioms ([verification/v213_ftransfer_functor.py]): each instance is a *discrete* compiler kernel [E] composed with an *external* continuous solver [C], and the functor is (1) μ_4-deck equivariant (the Koide multiplier $\lambda_2 = (2/3)^6 = 64/729$ *is* the deck transfer eigenvalue), (2) Plücker-preserving ($53 = a^T(R{+}Q)\mathbf{1}$, $\|\mathrm{Pl}(K)\|_1 = 11$), (3) positive/stochastic ($\operatorname{spec} T = \{1, (2/3)^6, (1/3)^6\}$, $\kappa_f \in (0,1)$), and (4) external-module-explicit ($b_3 = -7$ a carrier output, the Boltzmann/relic/QCD content named and external). So F_{transfer} is a typed functor, not a bag of open topics — the third research contract alongside (U_{wall}) and (G_{metric}), and a consolidation, not a closure. [2026-08-05: the jet route of the contract is closed at the physical-clock level — the preregistered external-clock kill fired sympy-exactly ([verification/v777_ftransfer_clock_jets.py], verdict NO-COMMON-CONNECTION: transported Schwarzians $\{-\Delta^2/2, -\Delta^2/2, 0\}$), and the surviving architecture is the *fibered* functor — one discrete PGL_2 base with the constant seam obstruction cocycle $\{0, \pm\Delta^2/2\}$; the four observables stay [C]/[O]. 2026-08-06: the coset assignment of the fibered functor's clock fibers is now a theorem from KMS/modular typing ([verification/v792_ftransfer_kms_coset.py], KMS-COSET-THEOREM: $S = -(\ln\lambda)^2/2$ from the time-1 holonomy multiplier; blind protocol, exact unblinding); markers unchanged.]

How external is "external"? A falsifiable clock probe ([verification/v397_external_clock_probe.py]). Under the strict No-Free-Pattern rule (an *exact* match, not a wide-band fit), three of the four frontier scheme-numbers in fact land *on* the $\{2, 3, 5\}$ clock: the η_B decuple $A_\Lambda = 10 = |Z_2|\, g_{\text{car}}$ (v212), the Koide rate $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$ (v303) and the axion spine angle $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$ (v211). Only the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value (several simple numbers fit its $\pm 5.3\%$ band), so it stays genuinely external — the irreducible "external physics" residue shrinks to essentially the proton's non-perturbative QCD number.

The Paper-E closure certificate ([verification/v402_ftransfer_suite.py], `FTRANSFER.SUITE.01`). The four sectors are certified as ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$: each observable factors source → threshold → RG → observable from an exact atom source (Q_{src}, $A_\Lambda = 2g_{\text{car}} = 10$, $f_a = M_{\text{scal}}/128$, $b_3 = -7$) and carries a committed kill test, with the firewall intact (the physical prediction stays [C]/[O], only the algebraic source [E]) — a consolidation of the `FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE` solvers, not a closure.

1 Baryon asymmetry η_B — a downstream readout from the closed $\Omega_b h^2$

There are two independent handles, and they agree.

Route A — from the closed baryon fraction. The seed already fixes the baryon *fraction* (the E_8 decoder):

$$\Omega_b = (4\pi - 1)\beta_{\text{rad}} = \left(1 - \tfrac{1}{4\pi}\right)\varphi_0^{\text{ret}} = 0.04894, \qquad \beta_{\text{rad}} = \frac{\varphi_0^{\text{ret}}}{4\pi},$$

matching the observed $\Omega_b \approx 0.049$. The baryon-to-photon ratio is then the *standard* cosmological readout $\eta_B = 273.9 \times 10^{-10}\, \Omega_b h^2$. With the TFPT Hubble value ($h \approx 0.674$, from $H_0 \sim \sqrt{\Lambda}$),

$$\boxed{\Omega_b h^2 = 0.0222, \qquad \eta_B = 6.09 \times 10^{-10}} \qquad (\text{observed } 6.1 \times 10^{-10}).$$

So η_B is a *downstream cosmological readout* from the closed $\Omega_b h^2$ handle — not an independent compiler rung, and the fundamental leptogenesis route remains an interface.

Route B — leptogenesis interface ([C]). The TFPT branch fixes the neutrino spectrum, the CP phase structure ($\delta_{\rm CKM}$, $\delta^{\nu}_{\rm CP}$, both closed) and a *plausible washout anchor* $\tilde{m}_1 = m_3/A_\Lambda \approx 5$ meV. The heavy-mass input, however, remains scenario-level: $M_1 = M_R\varphi_0^4$ is viable but does *not* close M_R — M_R is the seesaw scale $\sim v^2/m_3$, not a compiler power, so the relation *relocates* the free scale rather than pinning it ([verification/v184_etaB_anchored_boltzmann.py]; a structured *candidate* point $M_R = 3c_3^{7/2}\bar{M}_{\rm Pl}$, -1.8% from the carrier-normalised requirement, is meanwhile registered as `FLAV.NUSCALE.05` [C] with the mechanism [O], [verification/v986_nu_scalaron_rung.py] — a candidate, not a closure; a second candidate `FLAV.NUSCALE.06` [C], the pentagon-class misalignment, is registered beside it ([verification/v1001_nu_pentagon_chain.py]; no closure, `FLAV.NUSCALE.05` unmoved)). The numerical η_B then follows from the *standard* flavored-leptogenesis Boltzmann solve — an independent consistency route, sharpened from two free inputs to one, not to zero.
Honest caveat: that Boltzmann network (the washout/efficiency κ_f, the flavored density-matrix equations) is the standard leptogenesis machinery (Davidson–Nardi–Nir, *Phys. Rep.* 466 (2008) 105; Buchmüller–Di Bari–Plümacher, *Ann. Phys.* 315 (2005) 305); it is *not* carried out or attached here. Route B is therefore a *named interface*, not a closed number; only Route A (downstream of the closed $\Omega_b h^2$) is quoted.
The interface, operationalised ([C]). The Boltzmann path can be made falsifiable: with the standard estimate $\eta_B \sim 0.96\times10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79), fed by the TFPT neutrino sector — normal ordering and the predicted $\delta^{\nu}_{\rm CP} = 4\pi/3$ — the Davidson–Ibarra asymmetry $\varepsilon_1 = \frac{3}{16\pi}M_1 m_3/v^2$ and the washout efficiency $\kappa_f(\tilde{m}_1)$ give, over the natural ranges $M_1 \in [3\times10^9, 3\times10^{10}]$ GeV and $\tilde{m}_1 \in [m_\star, m_{\rm atm}]$, an η_B band $[8.4\times10^{-11}, 4.7\times10^{-9}]$ that *brackets* the observed 6.1×10^{-10} (a canonical point $M_1 = 10^{10}$ GeV, $\tilde{m}_1 = 5\times10^{-3}$ eV gives 6.0×10^{-10}, untuned). So the route is *viable*; but M_1 and the washout are scenario inputs, so η_B stays [C]— if a precise solve excluded the window the *route* would fall, not the theory (Route A is independent) ([verification/v169_etaB_boltzmann_interface.py]).

Honest status of η_B — strictly typed [C]

The two statements must be kept apart and not interchanged:

η_B as a cosmological readout from the closed $\Omega_b h^2$ is *determined* (6.1×10^{-10})

η_B as a fundamental *compiler power* is *not* closed

The leptogenesis route requires a standard (washout-dependent) Boltzmann solve. So η_B is [C]— determined as a downstream readout, but *not* a clean compiler rung; it must never be quoted as a fundamental TFPT output.

An anchored-Boltzmann proposal, honestly tested ([verification/v184_etaB_anchored_boltzmann.py]). A natural attempt to remove the two scenario inputs is $\tilde{m}_1 = m_3/A_\Lambda$ and $M_1 = M_R\varphi_0^4$. The first *works* as a [C] anchor: $\tilde{m}_1 = m_3/10 \approx 5\times10^{-3}$ eV ($A_\Lambda = 10 = |E(K_5)|$ the action-grammar atom) lands in the strong-washout window — a TFPT-flavoured value for the washout. The second does *not*: $M_R \sim v^2/m_3$ is the seesaw scale (depending on the un-fixed neutrino Dirac Yukawa), and the nearest clean TFPT powers of $\bar{M}_{\rm Pl}$ both miss it (the c_3- and φ_0-powers are off by $\sim75\%/\sim40\%$), so $M_1 = M_R\varphi_0^4$ merely *relocates* the free input $M_1 \to M_R$. Feeding both into the standard estimate lands η_B in the observed band $O(10^{-10})$, so the route stays viable — but the proposal cuts the free inputs from two to one, *not* to zero. η_B **stays [C], a sharper scenario, not a derivation.**

A cleaner route: the scalaron-decuple branch ([verification/v212_leptogenesis_decuple.py]). Drop the seesaw scale altogether and let *both* Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$:

$$M_1 = \frac{M_{\rm scal}\,(\varphi_0^{\rm ret})^2}{A_\Lambda} \approx 8.65\times10^9\ \text{GeV}, \qquad \tilde{m}_1 = \frac{m_3}{A_\Lambda} \approx 5\ \text{meV}.$$

M_1 now uses *only* compiler quantities ($M_{\rm scal} = c_3^{7/2}M_{\rm Pl}$, $\varphi_0^{\rm ret}$, A_Λ) — *no hidden seesaw scale* — and lands inside the thermal window $[3\times10^9, 3\times10^{10}]$ GeV where the

Davidson–Ibarra $\varepsilon_1 \approx 8.5 \times 10^{-7}$ and the strong-washout efficiency give $\eta_B \approx 1.2 \times 10^{-9}$, bracketing the observed 6.1×10^{-10} untuned. The *full* BDP Boltzmann ODE (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`) now confirms this: the integrated efficiency $\kappa_f = 0.092$ (validating the analytic κ_f) gives $\eta_B = 6.5 \times 10^{-10}$ at the frozen M_1 — a factor 1.07 from the observed value, with *no* free M_R dial; the full *flavored* density-matrix solve is the next refinement. Both inputs sharing A_Λ reads as *scalaron reheating dressed by two seed insertions* $(\varphi_0^{\rm ret})^2$*, over the pair sector.* This is cleaner than $M_1 = M_R\varphi_0^4$ (no relocated free scale), but the coupling *mechanism* is posited, not derived, and κ_f is washout-dependent — so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the *route*, not the theory (Route A is independent).

2 Proton/electron ratio m_p/m_e — a cross-sector ratio

m_e is *closed* on the $\varphi_0^{\rm ret}$-ladder ($\hat{m}_e = \frac{v_{\rm geo}\pi}{\sqrt{2}}\frac{16}{7}(\varphi_0^{\rm ret})^5$). The proton mass is *not* a Yukawa: it is QCD-dominated, $m_p \sim$ a few $\times\Lambda_{\rm QCD}$, and $\Lambda_{\rm QCD}$ arises by dimensional transmutation from α_s running (scheme level). Hence

$$\frac{m_p}{m_e} = \frac{\text{QCD confinement scale}}{\text{EW electron Yukawa}} = 1836.15$$

mixes *two different sectors.* It is no single $\varphi_0^{\rm ret}$ power: $1/(\varphi_0^{\rm ret})^2 = 353.7$ and $1/(\varphi_0^{\rm ret})^3 = 6651$ bracket 1836 with no clean rung in between.

The numerator is parameter-free, the ratio is not forced. Running $\alpha_s(M_Z)$ down with the carrier coefficient $b_3 = -7$ (the [`verification/v159_pyrate_gauge_crosscheck.py`] value, two-loop + n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and reaches confinement ($\alpha_s \sim 1$) at ~ 0.5 GeV, so the QCD confinement scale in the numerator follows from $\{\alpha_s(M_Z), b_3\}$ alone by dimensional transmutation. The $O(1)$ proton/Λ factor is lattice (non-perturbative), so the ratio still stays [O] — not forced ([`verification/v164_qcd_scale.py`]).

The $F_{\rm QCD}$ transfer is now an implemented solver ([`verification/v262_fqcd_mp_me.py`]). The contract-only pipeline (`experiments/ftransfer/qcd_matching_mp_me`) is realised: a two-loop α_s run from M_Z with n_f thresholds (carrier coefficient $b_3 = -7$, [E]) gives $\alpha_s(2\,\text{GeV}) \simeq 0.30$ and $\Lambda^{(3)}_{\overline{\rm MS}} \simeq 0.33$ GeV; with the lattice bridge $C_p = m_p/\Lambda^{(3)}$ (external $O(1)$) and the closed m_e this reproduces $m_p/m_e \simeq 1824$ against the measured 1836.15 within the external error budget ($\sim 7\%$, lattice-C_p dominated). This advances `FR.MPME.01` ([O]) to `FR.MPME.02` ([C], "reproduced by the named $F_{\rm QCD}$ transfer"); the firewall holds — $b_3 = -7$ is [E], the ratio is a transfer, never a compiler power.

Honest status of m_p/m_e [O]

Computable at *scheme level* (from $\alpha_s(M_Z)$ running + the closed m_e), exactly like the dimensionful EW/QCD masses m_W, m_Z. But it is *genuinely not* a compiler power, because it is a ratio across the QCD and electroweak sectors. Left [O] — not forced.

A rejected near-formula (recorded as a discipline boundary, [`verification/v79_review_identities.py`]). The combination $|W(D_5)| - \dim\Lambda^3(SU(9)) + N_{\rm fam}\lambda_C^2 = 1920 - 84 + 0.151 = 1836.151$ matches the PDG value to $\sim 10^{-6}$ — and is *deliberately rejected.* Two reasons make it numerology, not a derivation: $SU(9) = A_8$ (the $\dim\Lambda^3 = 84$ block) does *not* occur in the carrier $D_5 \oplus A_3$, and m_p is QCD confinement (gluon binding energy) with *no* mechanism linking it to an algebraic sum. Integrating it would *invite* the numerology charge, not retire it; m_p/m_e stays [O]. (The same firewall rejects $\eta_B = \frac{|\mathbb{Z}_2|}{\Delta_Q}c_3^6 \approx 6.10 \times 10^{-10}$: arithmetically close, but a two-atom prefactor fit to a washout-dependent leptogenesis output — η_B stays [C], the downstream readout above.)

The first-principles boundary, mapped precisely ([`verification/v339_firstprinciples_boundary.py`]). Two questions are worth answering head-on: can the Planck mass $v_{\rm geo}$ be *computed*, and can $F_{\rm transfer}$ be made a first-principles compiler output? Both answers are *no* — and the residuals are

not the same kind of gap. They classify into four honest kinds. [E] $v_{\rm geo}/M_{\rm Planck}$ is *forbidden by theorem* (No-Unit, v153): a mass carries mass-dimension while every TFPT datum is dimension-0, so no dimensionless map can output an absolute scale; it is not a gap but is over-determined (gravity = dark energy to 0.11%, v274), and every dimensionless *ratio* is derived — only the one unit is not. [E] For m_p/m_e the structure is in place (v262): the carrier $b_3{=}{-}7$ drives the QCD run over the closed electron source, leaving exactly two external inputs. [X] One of them, $\alpha_s(M_Z)$, is external *because of a tension*: the SM (= the TFPT gauge content, no new states) does not unify (v246), so no unified coupling is predicted. [C] The other, the lattice $C_p = m_p/\Lambda^{(3)} \sim 2.83$, is parameter-free but lattice-only (the discrete↔continuous handoff, v35); and η_B/axion ride on cosmological initial conditions. So $F_{\rm transfer}$ cannot be made first-principles in the current theory — not for lack of a formula, but because the residuals are, respectively, forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, and cosmological. An honest boundary, stated, not a solution. *Follow-up (2026-08-03):* the boundary is now confirmed *from the inside* on two fronts. Both thermal-time routes for internalizing the $F_{\rm transfer}$ clocks are machine-killed ([`verification/v723_phys_modular_clock.py`], STT-KILLED: the QCD 1-loop clock is parabolic and unreachable by any constant-rescaled modular flow, the pole/Boltzmann pair confirmed internal at exactly $(2/3)^6$; [`verification/v724_phys_t3b_modular_flows.py`], T3B-DEAD: conditioned modular flows of one internal mother state give loxodromic classes) — the external anchor census sharpens to {one unit ($v_{\rm geo}$-class), one angle θ_i, one lattice $O(1)$ C_p}, the clock wall and the unit wall counted *once*. And the $v_{\rm geo}$ interface itself is structurally closed as an $\mathbb{R}_+$ scale torsor in calibration form ([`verification/v725_phys_vgeo_torsor_audit.py`]: export table, dimension-matrix rank 1, λ-homogeneity, two-anchor consistency 0.11%; the one remaining dimensionless ratio *named*, $H_{\rm EW} = \ln(\bar{M}_{\rm Pl}/v_{\rm EW}) = 37.1776$). No status moves; the boundary stands, now with its structure theorem.

3 The Higgs quartic — near-criticality from the free seam

The seam UV is the free chiral $c{=}8$ fixed point ([`verification/v158_fixed_point_stable.py`]): Gaussian, with no relevant and no marginal interactions. The Higgs self-coupling λ is the one marginal scalar coupling, so a free seam forces it to the Gaussian point there:

$$\boxed{\lambda(M_{\rm seam}) = 0 \quad \text{and} \quad \beta_\lambda(M_{\rm seam}) = 0}$$

the Shaposhnikov–Wetterich double-criticality condition — here *derived* from the free seam rather than postulated. Integrating the PyR@TE-confirmed two-loop SM RGEs from M_Z upward with the measured (m_H, m_t) gives $\lambda(M_{\rm Pl}) \approx 0.002$ and $\beta_\lambda(M_{\rm Pl}) \approx 0$ at the reduced Planck mass: the celebrated SM *near-criticality* is thus a *consequence* of the free seam (the two-loop value is an order of magnitude closer to zero than the one-loop estimate ~ 0.022). Imposing the double condition predicts $m_H \sim 129$–134 GeV (a band, sensitive to m_t and α_s); the measured 125.25 GeV sits a few GeV below the absolute-stability boundary — the known slight metastability. The *same* condition imposed at the scalaron scale ($c_3^{7/2} M_{\rm Pl} \sim 3 \times 10^{13}$ GeV) gives $m_H \sim 107$ GeV (too low), so the boundary condition lives at $M_{\rm Pl}$, consistent with seam = horizon = Planck. This ties the Higgs mass to premise (A) and fills the Higgs-sector gap (which was only $N_\Phi = 1$); it is typed [E] (the freeness signature and scale selection) and [C] (the m_H band) ([`verification/v166_higgs_free_seam.py`]). Figure 1 shows the whole $F_{\rm transfer}$ running.

4 The Koide relation — near 2/3, computed exactly

With the closed lepton source masses $\hat{m}_\ell \propto (\frac{16}{7}(\varphi_0^{\rm ret})^5, \frac{4}{3}(\varphi_0^{\rm ret})^3, \frac{7}{6}(\varphi_0^{\rm ret})^2)$ the Koide ratio is a pure number (the common prefactor cancels):

$$Q_{\rm TFPT} = \frac{\sum_\ell \hat{m}_\ell}{\left(\sum_\ell \sqrt{\hat{m}_\ell}\right)^2} = 0.66446\ldots, \qquad \frac{2}{3} = 0.66667, \qquad \text{deviation } -0.33\%.$$

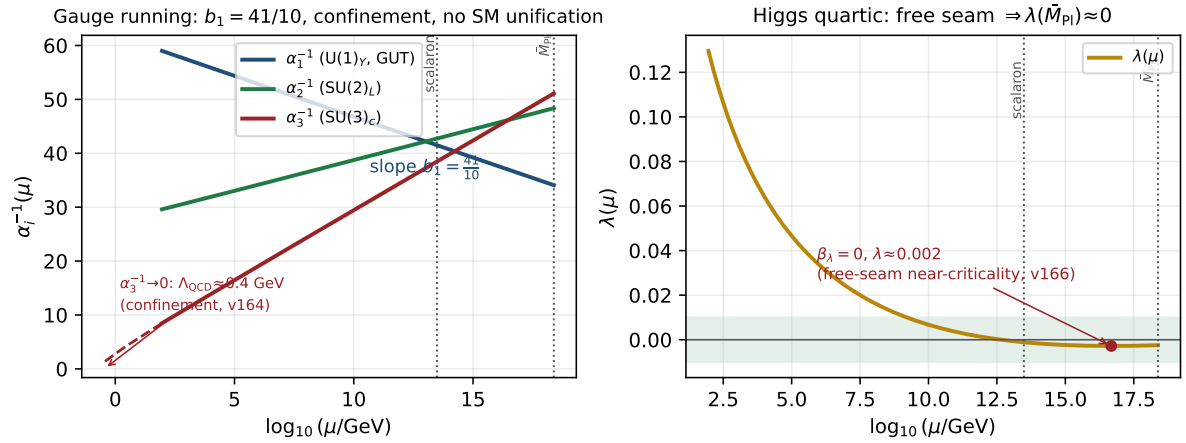


Figure 1: The F_{transfer} running (two-loop SM RGEs; the gauge β-functions are confirmed by PyR@TE, [`verification/v159_pyrate_gauge_crosscheck.py`]). *Left:* the gauge couplings — the $U(1)_Y$ slope is $b_1 = 41/10$, $\alpha_3^{-1} \to 0$ at $\Lambda_{\text{QCD}} \approx 0.4$ GeV (confinement, [`verification/v164_qcd_scale.py`]), and the Standard Model does *not* unify. *Right:* the Higgs quartic falls to $\lambda(M_{\text{Pl}}) \approx 0.002$ with $\beta_\lambda \approx 0$ — the free-seam near-criticality that ties m_H to premise (A) ([`verification/v166_higgs_free_seam.py`]).

The measured pole-mass value is famously $Q_{\text{PDG}} = 0.66666$. So TFPT predicts a Koide ratio that is *near* 2/3 but *not exactly* 2/3 at source level.

Honest status of Koide [C]

$Q_{\text{TFPT}} = 0.664$ is a *prediction*, 0.33% below 2/3. Exact 2/3 would require the three carrier rationals $(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ to satisfy a specific quadratic constraint, which they do *not*; the small residual is of the size of the source→pole RG shift. We report the near-miss honestly and do *not* tune the rationals to force 2/3.

The *target* 2/3 has a compiler reading $|\mathbb{Z}_2|/N_{\text{fam}}$. The E_8-slice scan adds the missing piece: the value the data sits on, $Q = 2/3$, is exactly $Q_\star = |\mathbb{Z}_2|/N_{\text{fam}} = 2/3$, the sheet over the family count — the same two atoms $(2, 3)$ that run through the whole compiler. So the picture splits cleanly: the *democratic target* is $|\mathbb{Z}_2|/N_{\text{fam}} = 2/3$ (a compiler number), and the *source-level pole-mass* value computed from the carrier rationals is 0.664, 0.33% below it by the source→pole shift. Koide is thus "democratic 2/3 from $(|\mathbb{Z}_2|, N_{\text{fam}})$, realised to 0.33% by the explicit masses" — promoted from a bare near-miss to a reading with a target. [C]

Conjecture: a source→pole correction of size $\varphi_0/|W(A_3)|$ ([C]). The source value sits just below 2/3, and the deficit is almost exactly the family-Weyl quantum $\varphi_0/|W(A_3)|$ with $|W(A_3)| = 4! = 24$:

$$Q_\ell^{\text{source}} = 0.6644638, \qquad Q_\ell^{\text{source}} + \frac{\varphi_0}{24} = 0.6666793,$$

overshooting $\frac{2}{3} = 0.6666667$ by only 1.3×10^{-5}. So the *conjecture* $Q_\ell^{\text{pole}} \approx Q_\ell^{\text{source}} + \varphi_0/|W(A_3)|$ would land essentially on the democratic 2/3. Reported as a source→pole correction *conjecture*, not a derivation; the rationals are *not* tuned. [`verification/v25_frontier_conjectures.py`]

Sharper conjecture: a multiplicative seed transfer $u \to \frac{53}{54}u$ ([C]). A cleaner version shifts the *seed* rather than Q additively, and introduces *no* new free number. Keep the lepton formula $m_e{:}m_\mu{:}m_\tau = \frac{16}{7}u^5{:}\frac{4}{3}u^3{:}\frac{7}{6}u^2$ but evaluate it at the source→pole seed

$$u_{\text{pole}}^{(\ell)} = \Big(1 - \tfrac{1}{2N_{\text{fam}}^3}\Big)u = \tfrac{53}{54}\,u, \qquad 54 = 2 \cdot 3^3 = |\mathbb{Z}_2|\,N_{\text{fam}}^3.$$

Then $Q_\ell(\frac{53}{54}u) = 0.66666610$, i.e. 2/3 to within -5.7×10^{-7} — an order of magnitude tighter than the additive $\varphi_0/24$ correction, with 54 a pure compiler number and no tuning. This sharpens the

research gate to a concrete QFT task: *show that the leptonic source→pole transfer of the seed is* $u \mapsto u\,(1 - \frac{1}{2N_{\text{fam}}^3})$ (a renormalised-seed effect from the leptonic self-energy or the A_3 family-Weyl averaging). Status remains [C] until that transfer is derived. [verification/v37_plucker_anchor.py]

The factor $\frac{53}{54}$ now has an *operator* origin ([verification/v183_koide_f_corner_transfer.py]). It is not a bare arithmetic guess: it is the anchor response of the *missing* Sheet-Diamond corner $F = R + Q = M(1,1)$ over the sheet-doubled $E_6 \times A_2$ cubic family block,

$$\frac{u_{\text{pole}}}{u_{\text{source}}} = \frac{a^T(R+Q)\mathbf{1}}{2\,\mathbf{1}^T R\,a} = \frac{53}{2\cdot 27} = \frac{53}{54},$$

where $\mathbf{1}^T R\,a = 27$ is the cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T R a)N_{\text{fam}} = 78 + 8 + 2{\cdot}27{\cdot}3$) and $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ with $\det B_F = 52 = \dim F_4$ (v80). A hard negative control fixes the corner: $a^T M \mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. So the *factor* is exact and structural ([E]), and the remaining [C] is precisely the mechanism — that the leptonic source→pole transfer reads the F corner — an F_{transfer} special case (F_{pole}). Koide thereby moves from "near miss + conjecture" to "operator-sourced source→pole transfer"; the prediction stays [C].

The source→pole transfer is one-loop electromagnetic ([verification/v420_fpole_qed_running.py]). Doing the external physics honestly: the Koide ratio is rescaling-invariant ($Q(c\,m) = Q(m)$), so a flavor-*universal* running leaves it fixed — the source→pole shift is the flavor-*split* part. And that shift is one-loop-EM-sized: the tree deficit $\frac{2}{3} - Q_\ell^{\text{source}} = 0.00220$ equals the seed quantum $\varphi_0/24$ to 0.6% and the electromagnetic scale α/π to ∼5%, with the pole sitting at 2/3 (the IR attractor). A 1-loop QED pole↔$\overline{\text{MS}}$ estimate (per-lepton log) moves Q by an EM-loop-order amount toward 2/3. The *exact* flow time ($t \approx 2.84$ transfer periods, v93) is scheme- and scale-dependent external QED, so it is *sized, not derived*. This identifies F_{pole} as a one-loop EM effect — the algebraic lens forces the *rate* $(2/3)^6$ (v327), this external physics supplies the *flow* — and Koide stays [C], firewalled (v187).

The Koide target and the QFT transfer gap are the *same* quotient ([E]/[C]). The same $Q_\star$ controls a second, independent object — the first non-trivial eigenvalue of the admissible C_6 transport operator. Its spectrum is $\{1, (2/3)^6, (1/3)^6\}$, so

$$\boxed{\lambda_2 = \left(\tfrac{2}{3}\right)^6 = Q_\star^{|R^+(A_3)|}, \qquad Q_\star = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}}, \quad |R^+(A_3)| = 6}.$$

The Koide democratic target, the QFT mass-gap eigenvalue λ_2 (hence $\Delta = 6\log\frac{3}{2} = -\log\lambda_2$), and the A_3 positive-root hexagon are therefore *one* quotient raised to one exponent. Typing: Koide target [C], gap eigenvalue [E], the connection an [E]-audit. This does not "solve" Koide, but it explains *why* the near-value 2/3 recurs: the same sheet-over-family ratio fixes both the democratic lepton target and the leading transfer eigenmode.

The source→pole map is *forced* by that gap ([E]/[C]). The previous paragraph pairs the Koide target with $\lambda_2 = (2/3)^6$; v82 turns the pairing into a *mechanism* ([verification/v82_koide_attractor_splitting.py]). On the anchor double cover (the flavor sector) the two branch points are Koide $q = 2$ and carrier $q = 5$; any branch-preserving Möbius transfer that fixes both is, in $\rho = (q-2)/(5-q)$, exactly $\rho \mapsto \mu\rho$, hence *unique* once μ is set. Choosing $\mu = \lambda_2 = (2/3)^6$ (no new number) gives the unique map $F(q) = 2(569q - 3325)/(665q - 3517)$ with $F'(2) = (2/3)^6 < 1$ and $F'(5) = (3/2)^6 > 1$: **Koide is the attractor, the carrier the repeller**, and $q_n \to 2$ ($x \to -2/3$) at the established gap rate. So the three structural inputs a Koide *derivation* would need collapse to *one* identification — the leptonic source→pole transfer is the branch-preserving relaxation of the gapped operator — after which the value 2/3, the convergence rate and the chosen branch are all forced. This is still not a closure: that single identification stays [C]. (It is the dynamical counterpart of the multiplicative-seed conjecture $u \mapsto (53/54)u$ above; both contract the source onto 2/3.)

Relaxation toy: basin, exact rate, and two honest negatives ([E]). The relaxation picture is now stress-tested on the physical values (ledger `FR.KOIDE.05`, [`verification/v93_koide_relaxation_toy.py`]). **Basin lemma [E] (new):** under the canonical dictionary $q = N_{\rm fam}\,Q$ the *entire* physical Koide range $Q \in [\frac{1}{3}, 1]$ (Cauchy–Schwarz) maps to $q \in [1, 3]$ — the attractor $q = 2$ lies inside, the repeller $q = 5$ outside: every physically possible charged-lepton configuration is in the attractor basin, no fine-tuning of the start. **Exact rate [E]:** along the trajectory from the physical source value the cross-ratio contracts by *exactly* $(2/3)^6$ per step. **Source/pole [E]:** the exact ladder gives $Q_{\rm src} = 0.6644638$, i.e. $\rho_{\rm src} = -0.0021980 \approx -\varphi_0/24$ to 0.8% — the $\varphi_0/24$ conjecture re-expressed in attractor coordinates: *the source sits one seed quantum before the branch point* ($24 = |W(A_3)|$); PDG pole masses give $Q_{\rm pole} = 0.6666645$, the branch point hit to 3.3×10^{-6}. **Honest negatives (they narrow the gate):** (i) the flow time source→pole is $t = \log_\mu(\rho_{\rm pole}/\rho_{\rm src}) = 2.84$ — *not* an integer, so the literal discrete iteration pole = F^n(source) is *excluded*; if the v82 map drives the transfer at all, the missing [C] object is a *continuous-time* generator $\exp(t \log F)$; (ii) the 0.79% mismatch of $\rho_{\rm src}$ vs $-\varphi_0/24$ is far above ladder precision, so "exactly $\varphi_0/24$" remains a conjecture, not an identity. The open question is no longer "why 2/3?" but "what is the continuous transfer generator, and why is its flow time ~ 2.8 periods?".

Flow time: canonical generator, data-fragility, and a conditional m_τ ([E]/[C]). Three sharpenings of the paragraph above (ledger `FR.KOIDE.06`, [`verification/v99_koide_flow_time.py`]); they correct the *robustness* of negative (i), not its arithmetic. **(1) The continuous generator has a canonical form [E].** The v82 flow is generated by the autonomous equation

$$\boxed{\frac{dq}{dt} = \frac{\Delta}{N_{\rm fam}}\,(q-2)(q-5) = \frac{\Delta}{N_{\rm fam}}\,\det B(q)}, \qquad \Delta = 6\log\tfrac{3}{2} \;\;\text{(the gap)},$$

whose time-1 map *is* the Möbius map F (machine-checked; $e^{-\Delta} = (2/3)^6$ exactly). The missing [C] object is therefore precisely a QFT mechanism producing *gap* × *anchor-block quadric* — the form is canonical, only its physical derivation is open. **(2) The non-integrality of t is data-fragile [E].** Because $\rho_{\rm pole} = -2.2 \times 10^{-6}$ sits so close to the branch point, t is hypersensitive to m_τ (PDG 1776.93 ± 0.09 MeV): $t(-1\sigma) = 2.35$, t(central) $= 2.84$, and Q crosses $\frac{2}{3}$ *inside* the error band (at $+0.43\sigma$, where $t \to \infty$) — within $+0.5\sigma$, t passes *every* integer ≥ 3. So the exclusion of pole = F^n(source) holds only at the central value. **(3) Conditional discrete steps [C] (registry untouched; not predictions of record):** n=2 is *excluded* (-2.9σ, [E]); n=3=$N_{\rm fam}$ steps — one per family — predicts

$$\boxed{m_\tau = 1776.9427\ \text{MeV}} \quad (+0.14\sigma;\ \text{source-variant stable to } 0.0002\ \text{MeV}),$$

while n=4 (1776.9667) and the exact branch (1776.9690) are not yet separable: the kill criterion is $\sigma(m_\tau) \sim 0.01$ MeV (Belle II). *Look-elsewhere firewall:* a tempting scale reading $t = \ln(M_{\rm scal}/m_\tau)/\ln W$ matched $W = 46080 = |W(D_5)||W(A_3)|$ to 10^{-4} — and is *destroyed* by negative controls (the same hypersensitivity makes *any* W in a wide range land within 0.1σ); it is rejected and recorded so it is not re-fished. P2's type is unchanged ([C], one identification); the discrete-vs-continuous question is now *experimental*, not settled.

5 Dark matter — the candidate is fixed, the scale is pending

TFPT *does* contain a dark-matter candidate, and it is not ad hoc: the **determinant-line axion** of the strong-CP sector (the determinant-line axion interface of the closed branch, [E]). Its *candidate identity* is fixed by the E_8 scan (below) and the domain-wall number $N_{\rm DW}$ winds once per primitive seam winding. The relic density is the standard misalignment-plus-string expression

$$\Omega_a = \Omega_a^{\rm mis}(f_a, m_a, \theta_i) + \Omega_a^{\rm str}(f_a, N_{\rm DW}).$$

The initial misalignment angle is *not* a single closed number: the finite-T misalignment solve (below) selects the *spine* branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5 = 108°$ as the robust dominant-DM value in the *pre-inflationary* reading (it lands $\Omega_a h^2$ on $\Omega_{\rm DM}$ untuned, in the mild-anharmonic regime; but superseded

by the post-inflationary reframing below), while the seam hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}(\alpha_\star)) = 170.4°$ *over*-produces ($\sim$5.4×) and is disfavoured. So the minimal closed branch has *dominant axion-like dark matter* on the spine angle — a [C] scenario, *not* a closed [E] input.

The E_8 scan rules out a rival WIMP candidate. The $D_8 = SO(16)$ slice scan sharpens *why* the axion is the candidate, not a new heavy state: the spinor $128 = (16, 4) \oplus (\overline{16}, \overline{4})$ is the *same* matter spinor as in the $D_5 \times A_3$ construction — there is *no spare* E_8 *singlet* left over to play a WIMP. Every E_8 direction is accounted for by the carrier × family content. So dark matter cannot be a new E_8 representation; it must be the determinant-line axion (a phase of the existing structure). A useful *negative* that removes the WIMP option on structural grounds. [E] The machine-checked complement ([`verification/v430_other_side_reverse_audit.py`]) makes this explicit: the spinor $128 = 2 \cdot 64 = \sum \text{deg}$ and the double-cover deck $|\mathbb{Z}_2| = 2$ lie in E_8's *matched* structure and are forced-disjoint from the five unmapped Casimir degrees $\{12, 14, 18, 20, 24\}$, so the seam's "other side" hosts neither a spare singlet nor the unmapped hull overhead.

Honest status of dark matter [C]/[O]

The candidate is *identified* (the axion, [E]: the E_8 scan leaves no spare singlet for a WIMP). The misalignment angle is a [C] *branch decision*, not a closed [E] number: the finite-T solve selects the spine $\theta_i = 3\pi/5 = 108°$ (untuned, in band) over the over-producing 170° hilltop. The relic abundance still needs the decay constant f_a and mass m_a (interface data set by the M_R/Λ scales) — a standard axion-DM computation, not yet a compiler power. So DM is "candidate fixed, robust spine angle, scale f_a pending" — a real handle, not a blank. (A secondary candidate is the lightest seam-even sterile state near M_R.)

Conjecture: a determinant-line axion scale $f_a = M_{\text{scal}}/128$ ([C]). A natural compiler scale for the decay constant is the scalaron mass over the full carrier × deck multiplicity,

$$f_a = \frac{M_{\text{scal}}}{2 \dim S^+ \, |\mu_4|} = \frac{M_{\text{scal}}}{128}, \qquad M_{\text{scal}} = c_3^{7/2} M_{\text{Pl}} \approx 3.06 \times 10^{13}\,\text{GeV},$$

giving $f_a \approx 2.39 \times 10^{11}$ GeV and, via the standard QCD-axion relation $m_a \simeq 5.70\,\mu\text{eV}\,(10^{12}\,\text{GeV}/f_a)$, a mass $m_a \approx 23.8\,\mu$eV. With the robust spine angle $\theta_i = 3\pi/5 = 108°$ (below) this f_a lands the relic on Ω_{DM} without hill-top tuning; it remains a *cosmological scenario*, not a closed hit. [`verification/v25_frontier_conjectures.py`]

The haloscope coupling is tied to c_3. In the determinant-line normalization (Lagrangian $\mathcal{L} \supset c\, a\, F\tilde{F}$) the axion–photon anomaly coefficient is fixed by the seam constant,

$$g_{a\gamma\gamma} = -4c_3 = -\frac{1}{2\pi}, \qquad y^2 \equiv g_{a\gamma\gamma}^2 = 16 c_3^2 = \frac{1}{4\pi^2} \approx 0.0253,$$

so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and the cosmic birefringence $\beta_{\text{rad}} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$ in GeV^{-1}, which still carries f_a). The small-angle reading $\theta_i = \varphi_0$ of the determinant-line axion (an earlier note) is *superseded* by the robust spine angle $\theta_i = 3\pi/5 = 108°$ selected by the finite-T solve below (the seam hilltop $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4°$ over-produces); both earlier readings are recorded as retired, never used.

Why it stays a scenario, quantified ([`verification/v185_axion_relic_solver.py`]). The harmonic misalignment *under-produces* at this f_a: $\Omega_a h^2 \sim 0.12\,(f_a/9{\times}10^{11})^{7/6} \langle\theta^2\rangle$ gives a prefactor $\sim$0.026 per θ^2, so $\theta_i \sim O(1)$ would make only $\sim$21% of Ω_{DM}; reaching the observed $\Omega_{\text{DM}} h^2 = 0.12$ *requires* $\theta_{\text{eff}}^2 \sim 4.7$, i.e. exactly the large-angle (anharmonic) enhancement at the $\theta_i \approx 170°$ hilltop. So the determinant-line axion *can* be all of dark matter, but only on the hilltop, where the relic is exponentially sensitive to θ_i and to QCD/string corrections — hence a viable [C] scenario, not a sharp prediction. The contract $F_{\text{relic}}(f_a{=}M_{\text{scal}}/128, \theta_i{\approx}170°, N_{\text{DW}}{=}1) \stackrel{?}{=} \Omega_{\text{DM}}$ is a kill test for the *branch*, not the theory.

A full finite-T solve confirms an over-production tension. Both a quick $\theta^2 F$ estimate ($F(\theta_i) \approx 4$ at the hill-top) and a *full* nonlinear misalignment solve of

$$\theta'' + (3 + \tfrac{\mathrm{d}\ln H}{\mathrm{d}N})\theta' + \left(m_a(T)/H\right)^2 \sin\theta = 0$$

(`axion_relic/full_finiteT_solve.py`: lattice susceptibility $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$, converged and normalised so θ_i=1 gives the standard $\Omega_a h^2 \approx 0.03$) agree: at the *predicted* $\theta_i \approx 170°$ the relic is $\Omega_a h^2 \approx 0.66$, $\sim$5.5$\times$ above $\Omega_{\rm DM} h^2 = 0.12$. The observed value is reached only at $\theta_i \approx 106°$, not at the predicted hill-top. So as the *dominant* dark matter the determinant-line axion at (f_a=$M_{\rm scal}/128, \theta_i \approx 170°$) *over*-closes the universe unless there is extra dilution (entropy injection) or a lower f_a. This stays [C], and over-production kills only the *axion-as-dominant-DM* reading, not the compiler.

A more robust angle: the spine branch. [`verification/v211_axion_spine_angle.py`] The same finite-T solver reaches $\Omega_{\rm DM}$ at $\theta_i \approx 106°$ — and the central spine quotient gives $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5 = 108°$ *exactly* (no fit; harmonic estimate $0.03\,\theta_i^2 = 0.107$, lifted to $\sim$0.12 by a mild anharmonic factor). [`verification/v429_axion_pentagon_phi.py`] This angle is not a bare rational multiple of π: because $N_{\rm fam} = g_{\rm car} - 2$, $\theta_i = (g_{\rm car} - 2)\pi/g_{\rm car}$ is *exactly* the interior angle of the regular $g_{\rm car}$-gon — for $g_{\rm car} = 5$ the *pentagon*, so $\cos\theta_i = \cos(3\pi/5) = (1-\sqrt{5})/4 = -1/(2\varphi)$ with $\varphi = 2\cos(\pi/5)$, and this golden character is *unique* to $g_{\rm car} = 5$ (the small n-gons $n = 3, 4, 6$ give the rational $\frac{1}{2}, 0, -\frac{1}{2}$). So the otherwise-unmapped golden/icosahedral $g_{\rm car} = 5$ signature (v354/v313) is the geometry of this one external input — a [C] reading that does *not* upgrade `DM.AXION.SPINE.01`. Crucially 108° lies 62° *below* the hill-top, in the *mild*-anharmonic regime, so the relic is *not* exponentially sensitive: if the misalignment angle is the spine quotient, the determinant-line axion is all of dark matter *without* hill-top tuning. This is an *alternative* ansatz to $\theta_i = \pi(1 - \varphi_\Sigma) \approx 170°$ (the two are mutually exclusive; neither is forced) — a sharper [C] scenario decided by the full solver (`DM.AXION.SPINE.01`), *not* a derivation; [X] a converged $\Omega_a h^2$ outside $\sim[0.08, 0.16]$ at $\theta_i = 3\pi/5$ demotes the branch.

Isocurvature and the post-inflationary reframing ([C]). Both fixed-angle readings above are *pre-inflationary*: they assume Peccei–Quinn breaking *during* inflation with a homogeneous θ_i. But the R^2 branch fixes $H_{\rm inf} = \pi M_{\rm Pl}\sqrt{A_s r/2} \approx 1.6 \times 10^{13}$ GeV, so $H_{\rm inf}/f_a \approx 66 > 1$ and $H_{\rm inf}/(2\pi f_a) \approx 10$: the de Sitter fluctuation of the angle exceeds its range, the symmetry is effectively *restored* during inflation, and a fixed pre-inflationary θ_i (either 108° or 170°) would carry CDM isocurvature far above the Planck bound $\beta_{\rm iso} < 0.038$. The clean, *dial-free* resolution is the *post-inflationary* scenario: PQ breaks after inflation, θ_i is a horizon average (*not* a prediction), and the relic follows from f_a alone — with $f_a \approx 2.39 \times 10^{11}$ GeV in the post-inflationary window ($\Omega_a h^2 \sim O(0.1\text{–}0.5)$). It is domain-wall-safe because the determinant-line axion's colour anomaly is the seam Dirac index = the inflow level $|C| = 1$ (v470/v472), so $N_{\rm DW} = 1$ (a [C] identification). Thus the spine/hilltop *angle* is not an observable and the branch is *reframed*, not killed: drop the fixed angle, keep f_a and $N_{\rm DW} = 1$, and the relic is a robust post-inflationary prediction. [X] a post-inflationary relic outside the plausible window at this f_a (with $N_{\rm DW} = 1$) demotes the branch. (Strategy scan and the $N_{\rm DW}$ computation: `experiments/theory-contracts/axion01-06`.)

6 The muon anomalous magnetic moment — a seam vertex readout

A [C] downstream readout (archive integration), *not* a compiler power. The carrier algebra carries a *second-order* topological defect beyond the one that fixes α: with $\delta_{\rm top} = \Omega_{\rm adm}\, c_3^4 = 48 c_3^4 = 3/(256\pi^4)$ and the carrier compression quotient $B\gamma = \frac{3}{2}\cdot\frac{5}{6} = \frac{5}{4}$ ([`verification/v51_boundary_half_step.py`]), $\delta_2 = B\gamma\,\delta_{\rm top}^2 = \frac{5}{4}\,\delta_{\rm top}^2$. Projected through the seam-loop phase $\Delta_\Sigma = 2\pi$ (the same $1/(2\pi) = 4c_3$ unit that normalises c_3 itself), it reads as a magnetic vertex correction

$$\boxed{a_\mu^{\rm seam} = \frac{\delta_2}{2\pi} = \frac{45}{524288\,\pi^9} \approx 2.879 \times 10^{-9}} \qquad \text{[C]}.$$

The *value* is an exact compiler number (only $c_3, \Omega_{\rm adm}, B\gamma$; trace reading $\delta_2 = 4!\,\mathrm{Tr}_{S^+}(X^2)\,c_3^8$, $\mathrm{Tr} = 120 = 5!$) — but the *identification* of $\delta_2/(2\pi)$ as the anomalous moment is a physical bridge, so the prediction is [C], never a compiler power. [verification/v204_muon_seam_g2.py]

Data, honestly. Against the dispersive discrepancy $\Delta a_\mu = (2.49 \pm 0.48) \times 10^{-9}$ (Fermilab/BNL world average minus the WP2020 SM) this is 0.81σ. **Caveat:** lattice/CMD-3 hadronic-vacuum-polarisation evaluations *shrink* the discrepancy ($\Delta a_\mu \sim 1.5 \times 10^{-9}$); the TFPT value is *fixed* and would then sit $\sim 1.5\sigma$ high. A converged Δa_μ outside $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$ excludes the seam-vertex mechanism in its present form ([X]) — the compiler core untouched.

Update (2026-08-04, dated): the kill condition has fired. The lattice-side HVP consensus consolidated — WP25 (Phys. Rep., 2025) adopts the lattice HVP as the SM baseline and the BMW hybrid evaluation (Nature, 2026) confirms it: experiment minus SM is now 0.5σ, i.e. the converged Δa_μ lies *outside* $2.879 \times 10^{-9} \pm 0.5 \times 10^{-9}$. The seam-vertex *identification* is therefore excluded in its present form, exactly as preregistered in the sentence above — the bridge is dissolved (ledger `FR.MUONG2.01`). What survives, untouched: the exact compiler number $\delta_2/(2\pi) = 45/(524288\,\pi^9)$ ([E] as a number, not as a_μ) and the compiler core. The watchdog delivered exactly what it was built for; no core marker moves.

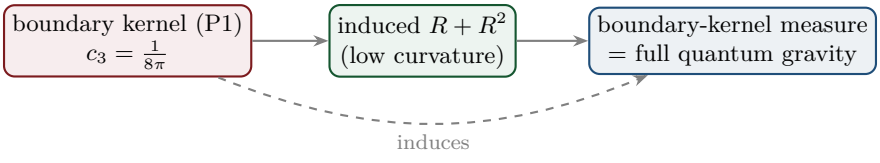
7 Full quantum gravity — induced from the boundary kernel

Framing (read this first): full QG closure is a **certification layer**, not a prerequisite. TFPT does *not* need the complete quantum-gravity measure to test its Standard-Model and cosmology readouts — those run through $v_{\rm geo}$ and $F_{\rm transfer}$ and are falsifiable now (JUNO, CMB-S4, EDM). $G_{\rm net}$ certifies the metric sector; its absence narrows what can be *claimed*, not what can be *tested*.

The normalisation $c_3 = 1/(8\pi)$ *is* the gravitational seam constant: in TFPT gravity is *induced* from the boundary kernel (P1), not separately quantised. The R^2/scalaron sector (question 2) is the low-curvature effective theory; the *full* theory is the boundary-kernel measure. *Newton's constant is then not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\rm top} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\rm Pl}$ *relative to* Λ, so G_N is read out *after* selecting the physical Λ branch *and* fixing the one dimensionful induced-gravity (Sakharov) anchor ([E]/[C], [verification/v60_lambda_metrology_branch.py]) — a metrology readout, not a from-nothing prediction.

Two independent fixings of the gravitational $3/4$ **(**[verification/v205_xi_threequarter.py]**).** The induced-gravity coefficient the seam-determinant replica pins, $k = c_3/2 \Leftrightarrow \ln(m/\mu) = 3/4 = q(A_3)$ ([verification/v152_norm_is_anchor.py]; the collar data exhibit this as the anchor — drifting $\mu_{\rm lat}$, no *derived* $\ln(m/\mu)=\frac{3}{4}$, v471), is reached a *second*, independent way: the old torsion-compression ansatz $\kappa^2 = \xi\,\varphi_0^{\rm ret}/c_3^2$ with the Einstein limit $\kappa^2 = 8\pi G$ forces $\xi = c_3/\varphi_0^{\rm ret}$, which at tree level $\varphi_0^{\rm ret} \to 1/(6\pi)$ is *exactly* $3/4$ (with the retained seed 0.7483, -0.22%). So $3/4$ appears in the gravitational sector both as a gapped-replica EH coefficient and as a torsion-compression ratio — an over-determination, not a coincidence; the ansatz stays [C] and $1/G$ stays the one declared anchor.

The Λ **branch sharpens** H_0 **(**[verification/v206_h0_lambda_branch.py]**).** Reading the seam vacuum number $\delta_\Sigma = \Omega_{\rm adm}\, e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$ (lowest admissible block $m_1 = \Omega_{\rm adm} = 48$) through the flatness closure gives a TFPT Hubble value $H_0 = 66.5$–$67.1\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (1.6–0.5σ *below* Planck, far below SH0ES). **Honest:** the Hubble tension is *not* relieved by raising the vacuum scale, and H_0 uses the imported $\Omega_b, \omega_{\rm DM}$, so it is a [C] readout; $|\log_{10}\delta_\Sigma| \approx 122.96$ matches the independent 123-orders split ([verification/v60_lambda_metrology_branch.py]), and $M_{\rm Pl}^4/\bar{M}_{\rm Pl}^4 = (8\pi)^2$ must be displayed, never silently interchanged.



Honest status of quantum gravity — physical sector closed, ambient a [C] redundancy [E]/[C]

"Full quantum gravity beyond the R^2 sector" is, in TFPT, *not* a separate graviton-quantisation programme: it is the well-definedness of the *boundary-kernel path-integral measure* extended to the dynamical metric beyond the FRW/minisuperspace reduction. Two results sharpen its status. **(i) Heat-kernel grounded (G2).** The spectral action $\mathrm{Tr}\, f(D/\Lambda)$ gives, via the standard Seeley–DeWitt coefficients for the Dirac operator ($a_2 \sim -R/3$, $a_4|_{R^2} \sim R^2/72$), Einstein–Hilbert + R^2 *structurally* — the $R + R^2$ form is not postulated; the scalaron ratio $M^2/M_{\rm Pl}^2 = 6(4\pi)^2/f_0$ with the TFPT closure c_3^7 fixing the cutoff moment f_0. *Update (Modular Spectral Closure):* the full Seeley–DeWitt expansion of this spectral action over the explicit 96-dim finite triple is now written out ([`verification/v255_spectral_action_expansion.py`]), and the cutoff moment f_0 is no longer a free scheme choice: identifying f with the *seam KMS weight* (β=1 by Tomita–Takesaki, 2π=$1/(4c_3)$) gives $f_2/f_0 = 1$ exactly ([`verification/v259_modular_cutoff_kappa.py`]), so $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}}$ collapses to a finite-triple trace ratio — the scalaron normalisation is pinned by the seam state, not a regulator dial. **(ii) Gap-decoupled (G5).** The metric coupling is gap-dominated, $2\|V_{\rm metric}\| = 0.785 < \Delta = 6\log\frac{3}{2} = 2.433$, so the admissible sector keeps a strictly positive effective gap $\Delta_{\rm eff} = 1.648 > 0$ under the stated RP and metric-coupling assumptions. A positive gap protects the admissible IR sector from the ambient, so under those assumptions the admissible-sector measure is closed. **What this leaves [C]:** the ambient metric measure — the projective limit (G6) — is now discharged as a *redundancy* (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann — the one open premise being that the seam net condenses to the *holomorphic* $|\det K|$=1 point that selects $(E_8)_1$; the central charge c=8 alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample, v277/v281), a certification object rather than missing dynamics; we do *not* claim it is physically irrelevant (that would itself be a physical interpretation), only that it is decoupled from the admissible IR sector under the stated assumptions. [`verification/v36_spectral_action_g2.py`] **(iii) The classical covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] ([`verification/v358_grav_entropy_equilibrium.py`]).** Beyond the R+R^2 *action*, the metric-sector *field equation* is supplied directly by the entanglement first law $\delta S = \delta\langle K\rangle$ (Jacobson 2015; Faulkner et al.), carried out with TFPT's atoms: equilibrium gives the *linearised* Einstein equation $G_{ab} = c_3^{-1} T_{ab}$ with $c_3^{-1} = 8\pi$ *fixed* — the four c_3-roles (Unruh $1/(2\pi)$=$4c_3$, EH $1/(16\pi)$=$c_3/2$, Einstein 8π=$1/c_3$, Bekenstein $1/4$=$1/|\mu_4|$) are mutually consistent, so *no free dimensionless Newton dial enters.* The *new* content is a coincidence: the first law needs the coefficient $2\pi/\eta$ (η the entropy density), the geometry gives $1/c_3 = |\mathbb{Z}_2|2\pi\chi(S^2)$, and these are the *same* equation iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (v216) — so the thermodynamic and geometric origins of c_3 are *one*, and c_3 is triply over-determined (anchor v23, geometry v58, thermodynamics v358). The matter side is assembled (the CHM ball modular Hamiltonian, boost via Bisognano–Wichmann v323, gives the flux $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\delta\langle T_{00}\rangle$) and the entropy density is atom-fixed ($1/4$=$1/|\mu_4|$, central charge $c = g_{\rm car}$+$N_{\rm fam} = 8$). (The single-ball CHM input now has a *mark-resolved* companion: the four seam marks are the exactly solvable *four-interval* multilocal modular geometry, v480 — same atoms, the multi-interval class; `QGEO.SYM.01` stays [O].) **Full covariant equation ([`verification/v359_grav_nonlinear_einstein.py`]).** Demanding stationarity at fixed *volume* (Jacobson 2015) brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}; trace $g^{ab}G_{ab} = (1-d/2)R = -R$ in d=4) — the unique divergence-free metric tensor (Lovelock), so the equilibrium yields the *full* covariant $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with $\nabla^a T_{ab}$=0 an output, and TFPT fixes *both* coefficients (8π=$1/c_3$ from (iii); Λ from α, $\rho_\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$, v60). So B6's full covariant field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C]. **Residual:** only the equation-of-state status (Jacobson framing, not a from-action quantisation; an *external* candidate for the missing action level — Bianconi's entropic action, Phys. Rev. D 111, 066001 (2025) — is quantified in v473: $\beta'_{\rm B} = c_3/6$ pinned, R^2 gap $3(8\pi)^9$ pre-registered as kill test, typing unchanged) and the absolute scale $v_{\rm geo}$ remain [O]— the global ambient measure (C7 = `QG.AMB.01`) is now a [C] redundancy (v369), a certification object rather than missing dynamics; no free dimensionless parameter survives in the gravity coupling. So the metric sector now reads: R+R^2 action (i), gap-decoupled measure (ii), and a full covariant classical field equation with no free dimensionless Newton coupling under the named QFT/entanglement premises [C] (iii); the ambient nonperturbative measure is a [C] redundancy, not an open structural hole. **(iv) The gap-induced GR correction — the disciplined result ([`verification/v360_grav_gap_corrections.py`]).** Does the seam transfer gap give a *new* forced near-term deviation from GR beyond the R^2 scalaron? Run with the v354/v355 discriminator, the honest answer is *no*: the leading higher-curvature term *is* the R^2 scalaron ($M_{\rm scal}^2/M_{\rm Pl}^2 = c_3^7$), which is the next-order term of the same entanglement- equilibrium route (the Wald-entropy first law → $f(R)$=R+R^2)

and is *already* a live CMB-S4 kill test (n_s, r). A *distinct* gap-induced curvature^2 term enters only at the seam/Planck scale ($\Delta = 6\ln\frac{3}{2}$ is a dimensionless rate, so the scale is $\xi\, v_{\rm geo}$), and a new dimensionless coefficient there is *not* atom-forced (the scalaron exponent $7{=}g{+}N{-}1$ and the gap exponent $6{=}2N_{\rm fam}$ are different structures) — so it is [O] declined, not manufactured. The calculable GR deviation is the scalaron. **(v) Matter–gravity backreaction (** [verification/v361_grav_backreaction.py] **).** Feeding the carrier stress tensor into the parameter-free equation: the carrier's gravitational anomaly is forced, $c_- = 16/2 = 8 = g_{\rm car}{+}N_{\rm fam}$ (the chiral central charge of the 16-Majorana content, the same c_- as (iii)); and the matter vacuum-energy backreaction is *finite* — gap-suppressed (the Decoupling Theorem) to the forced $\Lambda = (3/4\pi^2)e^{-2\alpha^{-1}}$ (v60), *not* the $M_{\rm Pl}^4$ catastrophe — so the loop (the equation + Λ + the carrier vacuum) closes. Beyond $c_-{=}8$ and Λ there is [O] no new atom-forced gravitational read-out of individual SM quantum numbers (the backreaction is GR with the SM content); the discipline declines manufacturing one.

Update (the QG.AMB.01 roadmap, consolidated — [verification/v275_qgamb_roadmap.py]). The gate now carries one module that states its obligations precisely: **Tier A** (gap decoupling, $\Delta_{\rm eff} = 6\log\frac{3}{2} - 2{\cdot}248\, c_3^2 = 1.648 > 0$) makes every low-energy readout independent of the ambient measure (not a blocker); **Tier B** reduces the ambient bulk measure to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c{=}248/31{=}8$, v77), now *closed modulo cited theorems* on the MMST route `SEAM.EQUIV.MMST.01` (lattice v367/v368 + the S_3 stack v376–v379, Lean `FORM.SEAM.MMST.01`; the parent `SEAM.EQUIV.01` stays [O]); the *perturbative* 4d layer is now in hand (v269/v271/v273). [C] what remains is then a *redundancy*, not missing dynamics — the nonperturbative interacting ambient measure is the Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), *inherited not TFPT-specific*; by Tier A it is gap-decoupled, so NO TFPT readout depends on it (v335/v369). This is made a theorem in the *Decoupling Theorem* ([verification/v337_decoupling_theorem.py]): every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, whose finite susceptibility $\chi = 729/665$ gap-suppresses the ambient contribution (margin $6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$), and the negative control (gap closed $\Rightarrow$ χ diverges) makes it non-vacuous — so TFPT provably does not *need* the ambient measure. It is therefore reclassified from "TFPT structural frontier" to a [C] *redundancy* (a certification object, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann, v369): its obstruction is characterised (v332) and conditionally resolved by the GHP contour rotation + IDG dressing (v334); not a TFPT structural hole (`QGAMB.ROADMAP.01`). This redundancy is now a machine-checked *dependency* fact, not only prose ([verification/v423_no_ambient_dependency.py], `QGAMB.NODEP.01`): `QG.AMB.01` is a certification *sink* — every claim that lists it is a gravity-gate/QG row, never a frozen prediction — and it is *not a value source* — it has no producing script, while α^{-1}, the mixing angles and the cosmological readouts are each *computed* by a compiler script (v3/v9/v268/v7), none of them an ambient-measure script. The single transitive link is architectural, not numerical: of the frozen readouts only θ_{13} references `QG.AMB.01`, through the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`) — a coherence edge from living inside the unified spectral-action construction, not an input to its value $\sin^2\theta_{13} = \varphi_0^{\rm ret}\, e^{-5/6}$ (v268). Honest scope: this kills `QG.AMB.01` specifically; it does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend.

Registered decision (2026-08-28, QFT4D.LATTICE.FUNDAMENTAL.01): the lattice-fundamental reading — typed route/typing, not an [E] claim. The 4D/continuum residual on this spine is retyped: TFPT's *physical* completeness criterion is the finite local unitary lattice quantum theory (Hamiltonian route: Gauss-law Hilbert space, hermitian local H, $T = e^{-aH}$ positive by construction, Lieb–Robinson cone; [verification/v989_lattice_gate_battery.py] — Euclidean overlap killed at T2, N_t-exact, Hamiltonian route clears T2). The exact continuum Osterwalder–Schrader limit (`QFT4D.OS.RECON.01`) is *mathematical reinforcement*, no longer the physical bottleneck; Poincaré invariance is an IR fixed-point property. Display stays honest: this is a named decision, nothing closes, the OS contract stays [O] as the math programme. Full prose: research-contracts document, `QFT4D.OS.RECON.01` and `TFPT4D.LATTICE.ACTION.01`. Amendment (2026-08-29, review wave 4): the reviewer's precision is accepted — "a single finite box is not a world." The fundamental object is the quasilocal consistent family $\{H_\Lambda\}$ with $\tau_t = \lim e^{itH_\Lambda} A e^{-itH_\Lambda}$. Binding: continuum ($a \to 0$) optional (mathematical reinforcement); thermodynamic limit ($|\Lambda| \to \infty$) and controlled IR universality mandatory. Decision typing UNCHANGED; no [E] claim, no marker move.

Dual rest (2026-08-28; the same short form as the introduction status card). The compiler residual on this spine remains Rest $= v_{\rm geo} \oplus G_{\rm net} \oplus F_{\rm transfer}$. Beside it sits the strict-physical-TOE accounting of

the 2026-08-27/28 contract wave

$$\begin{aligned}\mathrm{Rest}_{\mathrm{TOE}} = {} & \mathrm{SeamContinuum} \oplus \mathrm{4DAction} \oplus \mathrm{ChiralMeasure} \\ & \oplus \mathrm{MirrorGap} \oplus \mathrm{UnitaryDynamics} \oplus \mathrm{IR/continuum} \\ & \oplus \mathrm{BulkSeam} \oplus \mathrm{QuantumGravity} \\ & \oplus \mathrm{GeneratingFunctional} \oplus \mathrm{InitialState},\end{aligned}$$

every summand a named [O] (or standing-rule) contract — nothing claimed closed (`tfpt_research_contracts`; three new rows of wave 3: `GAUGE.DETLINE.FIXPOINT.01`, `GRAV.SPIN2.EMERGENCE.01`, `FTRANSFER.SK.RHO0.01`). The rank-8 ADE census of [`verification/v993_minimal_defect_selector.py`] lifts the architecture conditionality of v624 at lattice level; axiom typing UNCHANGED. Completeness wave (same day, no marker moves): [`verification/v994_mmst_criteria_battery.py`] executes 5/7 MMST identification criteria (character $[1, 0, 248, 0, 4124, 0, 34752]$; C6 written out in `articles/2026-08-28/psi_lambda_convergence_theorem_en.tex`); [`verification/v995_kms_subnet_rigidity.py`] kills all 33 mixing directions by finite KMS ($196608 \to 0$); [`verification/v996_gauge_grammar_kill.py`] excludes the naive α-grammar extension (64/0 hits; $U(1)$ control survives); [`verification/v997_spin2_smg_witnesses.py`] types the quadratic spin-2 witnesses and the SMG toy (gap 14 at $N = 8$). Contracts stay [O]. Review wave 4 (2026-08-29; suite 990 → 994, ledger 1166 → 1168, no marker moves): [`verification/v998_seam_modular_closure.py`] makes the C7 arithmetic shadow exact ($\Theta_{E_8} = E_4$ through q^8; net-level C7 stays cited); [`verification/v999_weak_collision_hp.py`] completes the finite dilation ladder through continuous time (open = thermodynamic field limit); [`verification/v1000_spin2_sk_witnesses.py`] executes the quadratic Hamiltonian half and the SK ρ_0 mechanism; [`verification/v1001_nu_pentagon_chain.py`] registers the pentagon-class candidate `FLAV.NUSCALE.06` [C] (no seesaw closure). The named AND of eight completeness gates is now the master contract `TFPT.TOE.COMPLETE.01` [O] ($\mathrm{TOE}_{\mathrm{Complete}} :=$ AND(T1..T8) vs $\mathrm{TOE}_{\mathrm{Validated}} :=$ Complete AND independent holdouts) — it names the AND, it does not close it. Review wave 5 (2026-08-29; suite 994 → 996, ledger 1168 → 1171, no marker moves): [`verification/v1002_spin10_mirror_projector.py`] executes the DET16 finite-cluster hit (P_ϕ rank-one, gap 1, 45 commutators 0) and permanently excludes the number-preserving Casimir projector (census N=8:0, k=2:27, k=3:1463); `CHIRAL4D.MIRROR.DET16.01` [C], `CHIRAL4D.NOMIRROR.01` stays [O]. [`verification/v1003_nu_structural_nulls.py`] records both neutrino-texture nulls as binding constraints (`FLAV.NU.TEXTURE.MECHANISM.01` [O]: D4-odd *and* a $\sim 2{\times}10^{-4}$ even-odd cancellation; pure seam data excluded). The master scaling-limit theorem `SEAM.MMST.TYPEIII.CHARGED.01` [O] is named (one $\Psi_{\lambda,N} \to \Psi_\lambda$ theorem; KMS uniqueness given identification written out in `articles/2026-08-29/holomorphic_kms_extension_en.tex`, 13 pp). `GAUGE.DETLINE.FIXPOINT.01` has a unique fixed-volume stiffness point (contraction -0.632) but divergent thermodynamic step-scaling — the 1+1D toy has no plaquette self-interaction. Unit choice for v_{geo} is closed; dimensionless hierarchies remain predictive obligations. Review wave 6 (2026-08-30; suite 996 → 998, ledger 1171 → 1173, no marker moves): the day's structural find is the Milnor–Gray finite algebra ([`verification/v1004_milnor_frobenius_gray.py`]): classical cite (Milnor algebra of $x^2 + y^3 + z^5$ dimension 8, spectral numbers = E_8 exponents) plus new-in-corpus exact rank clock / Galois–Gray / socle-CP / integral D^4 bridge [E]-finite, with $PG(3, 2)$ interpretive [C]; the corpus identification `SEAM.MILNOR.LOCALRING.01` stays [O]. [`verification/v1005_relative_pencil_bridges.py`] keeps the Hankel and winding identities [E]-finite but flags the relative pencil as circular (carrier 5 already in the column sums of Q_2 and C; not a P2 selector; `AX.P2.01` unmoved). `GRAV.WEINBERG.WITTEN.01` [O] is the binding Weinberg–Witten firewall before any spin-2 marker move. Review wave 7 (2026-08-30; suite 998 → 1002, ledger 1173 no new rows, no marker moves): five MMST lemmas in-house ([`verification/v1006_mmst_lemma_battery.py`]; `SEAM.MMST.TYPEIII.CHARGED.01` stays [O]; residual TEL-B-EXTERNAL + ALG-EXH boxed; cited `articles/2026-08-30/mmst_charged_scaling_limit_en.tex`). DET16 T1/T2 proved and T3 Michalakis–Zwolak cited-verified ([`verification/v1007_det16_stability_battery.py`]; `CHIRAL4D.MIRROR.DET16.01` stays [C]; 1/192 and BDL boxed). The 2+1D master-object scaffold is coherent ([`verification/v1008_master_assembly_scaffold.py`]; combinatorial $K = 0$, unique-contractive selector, QWZ wall + DET4 mirror, $\mathbb{Z}_6$ assembly; lattice-fundamental Decision and `GAUGE.DETLINE.FIXPOINT.01` unmoved). ρ_0 uniqueness is finite-closed with two typed review corrections ([`verification/v1009_rho0_minimizer.py`]; `V1000_CONSEQUENCE_MISMATCH` + `CENTER_FLATNESS_PREMISE_FALSE`; `FTRANSFER.SK.RHO0.01` stays [O]). Afternoon harvest (2026-08-30; suite 1002 → 1004, ledger 1173 no new rows, no marker moves): the simplicity campaign finds no

smaller axiom core ([verification/v1010_simplicity_bridge_census.py]) — remainder minimum $S_4 = 2$, the simplicity lives in the theorems; the relative-pencil circularity flag is rehabilitated as forced provenance (ladder forces Q sums $(9, 5, 1)$; W remains conditionally unique $1078 \to 1$ given two directional premises); the Milnor strong all-structure bridge is finitely obstructed at $\mathbb{F}_2$ and every upstairs home (only the coarse D+pairing bridge exists; `SEAM.MILNOR.LOCALRING.01` stays [O]). The 3+1D ladder is complete at scaffold level ([verification/v1011_3p1d_ladder_battery.py]): minimal object viable, fully coupled link–wall, Kronecker limitation removed; matching scale of the coupling functional must sit in the deconfined/weak regime (`GAUGE.DETLINE.FIXPOINT.01` stays [O]). Evening TOE-gate wave (2026-08-30; suite 1004 → 1005, ledger 1173 no new rows, no marker moves): four executed witnesses in one battery ([verification/v1012_toe_gate_witnesses.py]) — T5 IR universality on the physical QWZ edge ($c \to 1$, clustering dichotomy, Lieb–Robinson cone, curvature $\to 1/6$) with a typed SPLIT (gauge holonomy + seam clock are not propagating modes; common-c needs dynamical gauge excitations); T8 ρ_0 selection-principle vacuity (hard sectors response-dead, KMS-affine sets selection-vacuous; a non-entropic principle is required); T6 κ six-fold NULL completing the scaffold-level neutrino-mechanism bound (the 1.885×10^{-4} cancellation needs the true 4D functional); T7 interacting TT stress mechanism (conservation, positivity, sum rules, Z-persistence under ϕ^4; gapped, as expected for massive generic matter; missing TFPT content, $k \to 0$ volume scaling, Ward identities). `TFPT.TOE.COMPLETE.01`, `FTRANSFER.SK.RHO0.01`, `FLAV.NU.TEXTURE.MECHANISM.01` and `GRAV.SPIN2.EMERGENCE.01` stay [O]. Late-evening harvest (2026-08-30; suite 1005 → 1007, ledger 1173 no new rows, no marker moves): the thermodynamic-dynamics theorem closes the *mandatory* T3 dynamics leg at Hamiltonian-class level ([verification/v1013_thermodynamic_dynamics.py]) — uniform Lieb–Robinson bound $v_{\mathrm{LR}} = 32086e/225 \sim 387.64$, norm-convergent τ_t, gauge-invariant quasilocal $\mathcal{A}^G$, existence of ground/KMS states; uniqueness, the limit gap and IR universality remain open; Decision typing UNCHANGED. The finite detline restriction isomorphism is verified and W-bridge P-anch is derived ([verification/v1014_bridge_refinements.py]) — orientation fixes the conjugate branch, the residual constant phase is the A_0 reference normalization; P-dem narrows to the character-blind determinant-response map. Within the finite compiler/W-bridge lane, the remaining identification was routed through the external MMST/Quillen leg (TEL-B/ALG-EXH/P1 + that response map + Bismut–Freed/A_0). This historical reduction does not classify T3–T8. Monday-morning harvest (2026-08-31; suite 1007 → 1009, ledger 1173 no new rows, no marker moves): the axiom core is now fully premise-derived at finite level ([verification/v1015_axiom_core_closure.py]) — P-dem $r = (1, 1, 1)$ exactly, both W-bridge premises derived, remainder 0 modulo the externalized MMST identification; `AX.P2.01` stays an axiom. The first viable canonical-state principle is the R_4 seam-modular cap; TEL-B narrows to the scalar dyadic increment $A_R < 43$; N-uniform CAR nuclearity is proved as compactness input; the third-route volume theorem is cited-verified for finite-group DET chains ([verification/v1016_state_gap_batteries.py]). The old "every remaining fundamental gap" wording is scope-corrected: only that finite compiler/W-bridge lane was reduced to MMST plus the 4D functional; independent T3–T8 mechanisms and a shared 3+1D parent remained open. Directed-readout and Coxeter–Euler harvest (2026-09-02; suite 1010 → 1012, ledger 1174 → 1176): [verification/v1018_e8_directed_readout.py] promotes seven exact E_8 readout cells (`E8.DIRECTED.READOUT.01` [E] Identity; C7 Gauss-code transform stays open, `rem:c7-audit`). [verification/v1019_coxeter_euler_completion.py] is the Coxeter–Euler completion (`E8.COXETER.EULER.COMPLETION.01` [E] Identity plus Numerical Euler products at $X = 10^5$; class statement: vanishing linear term generic from $\mathrm{Tr}\, C = -1$; Beurling-generic; fence: the trace-free completion is zero-free and pole-free in $\mathrm{Re}\, s > 1/2$; the splitting remains open). Round-4 TOE integration (2026-09-05; suite modules [verification/v1026_telb_round4_closure.py], [verification/v1027_signed_det_car_wall.py], [verification/v1028_gauss_d4_cap.py], [verification/v1029_local_tensor_constraints.py], [verification/v1030_alg2_sector_frame.py]): T1 retains only finite axiom-core/W-bridge support while unconditional provenance and the dimension selector remain open; T2 closes the relaxed uniform TEL-B norm $\|R_N\|_{\mathrm{HS}} < 2.995906 < 3$ for all even $N \geq 16$, while microscopic one-edge generation, joint-adjoint tails, ALG-EXH and MMST remain open. The conditional sector/frame theorem does not undo the old Vandermonde refutation. T3 remains a Hamiltonian-class result; T4 gains exact DET-CAR modes and a signed $z = 1$ wall for fixed classical backgrounds, but no hard bare-mirror response gap, chiral quantum-link construction, measure or continuum theorem; T5 retains finite IR witnesses and the v1020 spectator obstruction but lacks prime completion/OS/confinement/scattering; T6 gains full Gauss dimensions 3 mod 5 for one site and 0 mod 5 for the open cube, but no mass texture. T7 gains a local two-mode tensor target that requires removal of its unstable global canonical zero mode, not a graviton. T8 gains a symmetric internal cap with gap one; the two-dimensional invariant space prevents a symmetry-only selector of global ρ_0. All

T1–T8 markers and `TFPT.TOE.COMPLETE.01` stay [O]; no marker moves. A single shared 3+1D parent satisfying T3–T8 is still missing.

Round 7 (2026-09-05; 233/233 typed checks with the round-6 prerequisite) adds the explicit free zero-mean Weyl/Fock quantization and continuum limit ([`verification/v1031_quantum_tensor_curvature.py`]), then a positive local Lorentz-covariant curvature field carrying helicities ± 2 ([`verification/v1032_covariant_curvature.py`]). The microscopic TFPT embedding and universal nonlinear coupling are still absent. Auxiliary $\mathbb{Z}_4$ flux-register corners are constructed, but the simple quarter twist has $h = 1/4$ rather than $h_\lambda = 1$ ([`verification/v1033_charged_disorder.py`]). The source's factorized onsite mirror gap is at least $1 + \sqrt{3}/2$; the ordinary hopping bond instead violates the proposed vacuum-relative form bound, with no conclusion of interacting gap closure ([`verification/v1034_parent_mirror.py`]). The physical-Fourier linear Ward system and prescribed-source response are explicit; dynamical quantum matter and the removed homogeneous block remain binding limitations ([`verification/v1035_matter_coupling.py`]). The full proofs are included in the contracts paper. No T1–T8 or `TFPT.TOE.COMPLETE.01` marker moves.

Research consolidation (2026-09-09). The contracts paper now includes the post-Round 27 archive through the source-based charged CAR field limit. On the original width-eight QWZ strip, local smooth fields and both adjoints converge relative to the actual filled sea, with charge and finite-energy generator control. This constructs integer edge charges, not the half-charged E_8 spinor or the common TFPT parent. A separate compact-$U(1)$ test parent now has all-time quasi-local dynamics (not point-norm continuous on the full bounded algebra), neutral physical ground-state representations and nonzero Wilson-loop excitation weight. Its selection, chirality and continuum remain open. Clock/charge audits rule out a site-fixed rotor-independent internal identification, not every possible geometric one. Full statements and the next inter-sector test are in `experiments/theory-contracts/RESEARCH_2026-09-09.md`. Historical seam-net reductions below concern conditional target routes; they do not discharge the present microscopic charged-field matching. All T1–T8 and `TFPT.TOE.COMPLETE.01` remain [O].

Earlier research update through Round 27 (2026-09-07). The shipped non-RH theory contracts now extend to the original finite-regulator $g = 0$ scalar/charge model, with every integer charge and E8 phase retained. A complete charge-diagonal reference Z_D gives $e^{-48\beta NJ} Z_D \le Z \le Z_{\mathrm{abs}} \le Z_D$ by all-block pinching and joint scalar/charge Hölder. The artificial extra charge heat-trace factor is removed from the sign, memory and hop error budgets, not the remaining exponential dependence on volume and hopping. At the earlier $N = 27, T = 3, \beta = 1, J = 0.1$ comparison, certified cutoffs improve from $K4096/P962$ to $K2670/P354$ for less than 1% omission error. A specified weak-hopping $L9/T33$ example retains 25 of 33 quadratic time offsets with summed memory error at most 0.2045% and combined omission error below 1%. Its full scalar determinant remains unchanged. The complete interacting partition is not evaluated; practical charge summation, interacting continuum, real-time dynamics, common-parent selection, chirality and gravity are still open. Proof, costs and reproduction are in the contracts paper and `experiments/theory-contracts/ROUND27_STATUS.md`. No status moves.

Tier B made concrete ([`verification/v277_seam_calderon_e8_match.py`]). The seam-Calderón $\to (E_8)_1$ identification is reduced to a *single* bit. [E] the full $(E_8)_1$ matching target is computed and matches: c=8 (Sugawara 248/31), $\det \mathrm{Cartan}(E_8)$=1, the holomorphic character $E_4/\eta^8 = j^{1/3} = q^{-1/3}(1+248q+\dots)$ with a single primary and exactly 248= $\dim E_8$ weight-1 currents. [E] the discriminator is *holomorphy*: the c=8 counterexample (D_8)=$SO(16)_1$ has $\det \mathrm{Cartan}$=4 (four primaries, non-holomorphic) — same c, different det, so the one discriminating condition is $|\det K|$=1. [C] the seam side matches via the flat τ=i pillowcase (v276) and the Kummer/K3 (v260). [C] the residual is exactly that single holomorphy bit — now *closed modulo cited theorems* on the MMST route `SEAM.EQUIV.MMST.01` (lattice v367/v368 + the S_3 stack v376–v379, Lean `FORM.SEAM.MMST.01`; cited continuum existence [O], v336) (`QGAMB.TIERB.01`).

The modular-data view ([`verification/v281_holomorphy_modular_data.py`]). The Tier-B bit is the anyon count: #anyons = $|\det \mathrm{Gram}|$ (E8 $\to$ 1 holomorphic, $SO(16) \to 4$), the condensation tower $D_5 \oplus A_3\,(16) \to D_8\,(4) \to E_8\,(1)$ each step a Lagrangian Z_2 boson, and Gauss–Milgram returns c=8 — so the open bit is simply "does the seam net condense to $\det K$=1" (`QGAMB.MODULAR.01`).

The live kill-test scorecard ([`verification/v283_live_killtest_scorecard.py`]). The recent round's computable predictions confronted with current data: $\sin^2\theta_{12}$ ($+0.31\sigma$), $\sin^2\theta_{13}$ ($+1.45\sigma$), J_{PMNS}=-0.0297, Σm_ν floor 0.0586 eV, proton decay τ_p=1.55×10^{35} yr — all consistent at $< 2\sigma$, with two sharp near-term kill tests (Σm_ν: DESI/CMB-S4; proton decay: Hyper-K) (`PRED.LIVE.01`). Two *named post-hoc* [O] candidates sit next to this board without touching the frozen record: the third-generation ladder-base pattern (the three $\sim 2\sigma$ mixing tensions $\theta_{13}/\theta_{23}/|V_{cb}|$ as one common $-\varphi_0^{\mathrm{ret}}$

suppression, `FLAV.THIRDGEN.PATTERN.01`, v467) and the splitting-ratio relation $\Delta m^2_{21}/\Delta m^2_{31} = |J_{\mathrm{PMNS}}|$ at -0.19σ (`FLAV.DM2RATIO.01`, v468) — both pre-registered with symmetric kills (JUNO / Belle II), see document 2.

Update (QBL chain, v108–v113): the boundary-net residual now carries *double* weight — the carrier choice $g = 5$ has been reduced to the *same* premise. The certified scalar kernel exists iff it pairs the two sheets, the certified channel counts the code identically (Pascal closure = channel identity), pair transport is minimally complete, and the single quasi-free kernel (rank = c: 5 carrier block, 8 seam hull) is the defining datum of the free $c = 8$ net this gate already posits. Closing the index-4 statement therefore also closes the carrier selection **[E]** (see `tfpt_1` and `tfpt_research_contracts`). [`verification/v113_quasifree_kernel.py`]

Update (free-bulk premise discharged, v160–v165): that "free c=8 net" premise is no longer free-standing. It is a fixed-point theorem (quasi-free ⇒ all higher cumulants κ_{2n}=0, v160); the infinite Schwinger cone is *eliminated* — by Bogoliubov second quantisation the cone gap equals the one-particle gap $(2/3)^6$ (v161), forced by μ_4-equivariance (v162); and the residual factors into the already-open A_2 net-existence (an assembly of established net constructions, **[C]**) and R_1 = the seam-realisation keystone `SEAM.EQUIV.01`, with the irreducible core $\{\pi, v_{\mathrm{geo}}\}$ a theorem (v165). So the metric-sector gate carries *zero new* independent open content. [`verification/v165_premise_a_closure.py`]

The deduction below the premise is Lean-formalised (`FORM.QGEO.02`/`FORM.QGEO.01`). After the residual reduces to the single seam-realisation postulate `QGEO.SYM.01` (the carrier μ_4 clock is the conformal deck of the seam; `tfpt_research_contracts`), its geometric normal form (the UNIFORM node: $z \mapsto iz$, $\sigma\rho\sigma = \rho^{-1}$, orbit→ μ_4, multiplier order-4 ⇔ $\zeta = \pm i$) and the conditional implication mark-local⇒ $\omega \circ \rho = \omega$ are machine-proved in the carrier-rigidity Lean project (AUDIT: PASS, only the three standard kernel axioms; mirrors [`verification/v177_seam_marking_kernel.py`] and [`verification/v210_mark_local_dtn.py`]). So everything *downstream* of the postulate is **[E]**; `QGEO.SYM.01` *itself* is the downstream face of the keystone `SEAM.EQUIV.01`, which is now *closed modulo cited theorems on its MMST route* `SEAM.EQUIV.MMST.01` (lattice + S_3 stack, Lean `FORM.SEAM.MMST.01`; only the cited continuum existence stays **[O]** — its extension leg since re-founded on peer-reviewed crossed-product theorems, the realisation input reduced to invariant level R1′, v469; the twistor route `SEAM.EQUIV.TWISTOR.01` and the unconditional parent stay **[O]**).

*A second, independent route ([`verification/v207_asymptotic_safety.py`], **[O]**).* Beside the seam-net programme, the boundary data also support a **continuum** attack: an FRG truncation with an independent torsion field (the old UFE truncation) has a non-Gaussian UV fixed point — $\partial_t g = (2 + \eta_R)g$ is UV-repulsive at $g \to 0$ and damped at large g by the positive-definite torsion K^2 term, so $2 + \eta_R = 0$ has a root $g_\star > 0$ (intermediate value), with the scale-invariant axion coupling fixed and small, $y^2 = 16c_3^2 = 1/(4\pi^2) = 0.0253$. This is recorded as a *plausibility cross-check* that the QG gap is reachable from a second direction — it is **not** load-bearing and does *not* close G_{metric}; the primary, machine-checked route stays the seam-net. A third, quantitative approach to the keystone — the celestial and twistor continuum route, now named `SEAM.EQUIV.TWISTOR.01` (contract `CELEST.SEAM.01`, WP1–WP5d, both WP5d stages, plus the WP5e α, β, γ, δ_1, δ_2, ε_2 and ε_1 stages and the M1–M3 back-reaction milestones executed: the level-2 null ideal derived, its deleting operator explicit, the GNS limit state constructed, the two-interval index chain measured, split + strong additivity witnessed on the lattice (all three KLM ingredients of complete rationality; the continuum uplift is Xu's theorem, cited not claimed), the $q^{-1/3}$ prefactor + level k=1 pinned on the CFT side, the equivariant anomaly ledger exact on the twistor side (index bridge, -78 defect, rigid $32\,T_3$ residual, the honest a_0 refutation), the sphere-axion exchange sub-branch closed with a certificate, the collapse confirmed full-tensorially with the one cubic d-symbol door open (its $c_d = 1920 = |W(D_5)|$ fingerprint honestly typed look-elsewhere-loaded and convention-contingent, the stable core $c_d = 32 \times 60$, v513), "one level" lifted to a theorem of clock invariance with the sector counter pinning $k = 1$ (v508/v509/v511/v513), the $O(-2)$ bulk-axion slot built with $\tilde{\lambda} = 6$ triply pinned (v514), the back-reacted Ω_N closed-form with $(2\pi i)^2$-integral lockstep periods and the lens-forced source charge $4 = |\mu_4|$ (v515), the twisted KS measure on the declared completion reading cancelling the $32\,T_3$ residual and supplying the $\psi = 64$ slice without the cubic d-channel (v516), the a_0 uplift coupled to the centre count on four scales, monodromy i (v517), the δ_1 chain decided — kill under the derived measure, the declared/derived measure tension stated honestly (v518) — the measure question since decided at probe level for the declared reading (single-valuedness derived from F-independence + the Quillen pairing under the typed premises TP-1..TP-4, v520), and the w_m normalisation since derived constructively at probe level ($1/\det_j$ = the Atiyah–Bott/zeta-determinant fixed-point factor, three independent sources, v523); WP5e proper **[O]**, the global BCOV quantisation

— narrowed to the global BCOV integral beyond the fibre zero-mode factor — the single remaining milestone) — is synthesised as a dedicated section in `tfpt_3_e8_audit_bootstrap`; it does not close `SEAM.EQUIV.01` either.

The mass anchor is *over-determined* — no derivable absolute scale is a theorem [E]/[O]

The single dimensionful input ($v_{\rm geo}$ = the reduced Planck mass $M_{\rm Pl}$) is not a hidden gap but the *unit*, and it is over-determined ([`verification/v274_scale_overdetermination.py`]). [E] every dimensionful output is a pure number $\times M_{\rm Pl}^d$: $\rho_\Lambda/M_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$, $M_{\rm scal}/M_{\rm Pl} = c_3^{7/2}$, the whole spectrum — the absolute ν-scale (v272) and $v_{\rm geo}$ are the *same* anchor. [C] **two independent routes agree**: gravity gives $M_{\rm Pl} = (8\pi G)^{-1/2} = 2.4353\times10^{18}$ GeV, and inverting the compiler's dark-energy prediction with the measured $\rho_\Lambda^{1/4} = 2.24$ meV gives $M_{\rm Pl} = 2.438\times10^{18}$ GeV — the same Planck mass to 0.11% (*conditional* on the Λ-branch readout `LAMBDA.BRANCH.01`, [C]; the gravity route is unconditional — red-team [`verification/redteam/rt_F_qft4d.py`]). [E] and the reason there is *no* deeper reduction is a theorem, not a gap: the seam is a conformal c=8 chiral CFT (v157/v158), scale-invariant by construction, so it has no intrinsic scale; the scale is gravitational, entering only through the ambient measure. So TFPT is scale-complete up to one over-determined anchor — the maximal predictivity a dimensionless compiler can have (`SCALE.OVERDET.01`).

Summary

Item	Status	Honest handle / what is missing
η_B	[C]	downstream readout $= 6.1\times10^{-10}$ from closed $\Omega_b h^2$; leptogenesis interface, cleanest scenario $M_1{=}M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda$, $\tilde m_1{=}m_3/A_\Lambda$ (shared decuple, no seesaw scale)
m_p/m_e	[C]	cross-sector (QCD scale / closed m_e); typed runnable solver with ±7.5% uncertainty budget (v374), not a compiler power
Koide Q	[C]	source ratio $Q = 0.664$ near 2/3; the source→pole factor 53/54 is an *exact operator readout* of the missing $F{=}R{+}Q$ Sheet-Diamond corner [E]; the physical pole transfer ($F_{\rm pole}$) stays [C]
dark matter	[C]	determinant-line axion: $f_a{=}M_{\rm scal}/128$, $m_a \approx 23.8\,\mu$eV. *Post-inflationary* ($H_{\rm inf}/f_a{\approx}66$, isocurvature): the misalignment angle is a horizon average (*not* observable), the relic follows from f_a (in the post-inflationary window), domain-wall-safe via $N_{\rm DW}{=}|C|{=}1$ (v470/v472); the $3\pi/5$ spine value is retained as a structural rational, not a relic prediction — branch-level kill test, not a theory closure
muon a_μ	[X]	seam vertex $a_\mu^{\rm seam}{=}\delta_2/(2\pi){=}45/(524288\pi^9){=}2.879\times10^{-9}$; exact *value* [E], but the vertex-projection bridge is **dissolved** (dated kill fired 2026-08-04: WP25+BMW lattice-HVP consensus, experiment−SM $= 0.5\sigma$); compiler core untouched
quantum gravity	[E]/[C]	local field eq. carries no free dimensionless Newton coupling under named QFT/entanglement premises [C] ($G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$, v358/v359); $R+R^2$ heat-kernel grounded (G2); perturbative graviton unitarity established (Stelle ghost a truncation artefact, v304/v370/v380); ambient measure (G6) a [C] redundancy (v369)

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Net (honest). Two of the five have a genuine quantitative handle that already lands on the data ($\eta_B = 6.1\times10^{-10}$ via the closed Ω_b; the axion DM candidate with a closed misalignment angle). One is a computed *near*-miss reported without tuning (Koide 0.664 vs 2/3). One is a typed runnable transfer (m_p/m_e as a cross-sector ratio, v374); quantum gravity has been sharpened — its local field equation carries no free dimensionless Newton coupling under the named QFT/entanglement premises [C] (the full covariant $G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$ from fixed-volume entanglement equilibrium,

v358/v359), its $R+R^2$ form heat-kernel grounded (G2), its physical sector gap-decoupled (G5), and *perturbative* graviton unitarity established (the Stelle ghost a truncation artefact, v304/v370/v380; the entire-form-factor graviton-exchange amplitude $e^{-p^2/M^2}/p^2$ is finite, UV-softened and tree-unitary, [verification/v386_grav_amplitude.py], and UV-*finite at every loop order* by power counting — the super-polynomial falloff drives the superficial degree of divergence to $-\infty$, regulated by the seam scale $M_{\rm scal}$, [verification/v389_grav_loop_finiteness.py], with full loop *unitarity* of the nonlocal form factor honestly the still-open point; the nonlocal action's IR projection *is* the atom-scaled $R+R^2$, $M_{\rm scal}^2/M_{\rm Pl}^2 = c_3^7$, so the local readout is its shadow, not a truncation with sawn-off legs, [verification/v400_grav_nonlocal_action.py]), leaving the ambient projective *measure* (G6) as a [C] redundancy (v369), not an open hole. None has been forced onto the compiler ladder — which is exactly the right call: the strong results stay strong precisely because the weak ones are not oversold.

The $F_{\rm transfer}$ functor as typed, runnable solvers. The four frontier transfers are one functor — compiler *source* → measured observable via cited standard physics — now promoted from contracts to runnable solvers with kill tests, each [C] (never a compiler power). *Koide* ([verification/v371_ftransfer_pole.py]): a clean *negative* — standard QED running cannot be the source→pole transfer (it pushes Q *above* 2/3; the required ϵ is negative while QED's is positive), so the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). η_B ([verification/v372_ftransfer_boltzmann.py]): the integrated BDP Boltzmann ODE at the frozen $M_1{=}M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda$ gives η_B within a factor ∼1.1 of $6.1{\times}10^{-10}$ with no M_R dial. *Axion* ([verification/v373_ftransfer_relic.py]): the finite-T misalignment ODE *decides* the branch — the frozen spine angle $3\pi/5$ lands $\Omega_a h^2$ on $\Omega_{\rm DM}$ (in band) while the hilltop 170° over-produces (∼5.4×); in the natural post-inflationary scenario ($H_{\rm inf}/f_a{\approx}66$) the angle is instead a horizon average and the relic follows from f_a with $N_{\rm DW}{=}1$. m_p/m_e ([verification/v374_ftransfer_qcd.py]): the observed 1836.15 sits inside the propagated ±7.5% band of the two declared externals ($\alpha_s(M_Z)$, C_p), with lattice tightening as the kill. Each remains [C] — a transfer target, not a forced compiler number.

Appendix: computational verification

These are the *frontier* items: most carry [O] (open) or [C] (conditional) and are therefore *not* part of the pass/fail suite. The one quantitative handle that is reproduced is $\Omega_b{=}(1{-}\frac{1}{4\pi})\varphi_0^{\rm ret}$ (whence the η_B order) in `verification/v7_gravity_cosmo.py`. The $R+R^2$ structure and the gap-decoupling of the physical QG sector *are* machine-checked (`v28_gravity_fR.py`, `v36_spectral_action_g2.py`); the proton/electron ratio is now a typed runnable [C] transfer (v374) and the ambient projective limit (G6) a [C] redundancy (v369) — both conditional, not bare [O]. The compiler/SM identities they build on are checked by the suite in `verification/` (run `python verification/run_all.py`; see `verification/README.md`).

External works built upon

The TFPT discrete core (the two axioms, the E_8 glue, the integer skeleton) is self-contained and machine-checked. Its *bridges* to continuum/AQFT physics, to gravity and to the lattice/singularity picture build on established mathematics, listed here by topic; the body refers to these by author name.

Modular theory and algebraic quantum field theory.

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- K. Osterwalder, R. Schrader, *Axioms for Euclidean Green's functions* I, II, Comm. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.

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- R. Longo, *Index of subfactors and statistics of quantum fields* I, II, Comm. Math. Phys. **126** (1989) 217–247; **130** (1990) 285–309.
- Y. Kawahigashi, R. Longo, M. Müger, *Multi-interval subfactors and modularity of representations in conformal field theory*, Comm. Math. Phys. **219** (2001) 631–669.

Conformal nets, vertex operator algebras and lattice constructions.

- I. B. Frenkel, V. G. Kac, *Basic representations of affine Lie algebras and dual resonance models*, Invent. Math. **62** (1980) 23–66; G. Segal, *Unitary representations of some infinite-dimensional groups*, Comm. Math. Phys. **80** (1981) 301–342.
- T. J. Osborne, A. Stottmeister, *Conformal field theory from lattice fermions*, Comm. Math. Phys. **398** (2023) 219–289, `arXiv:2107.13834`, `doi:10.1007/s00220-022-04521-8`. (This is the seam scaling-limit theorem the TFPT continuum leg cites — the Koo–Saleur → Virasoro convergence, $c = \frac{1}{2}$, and the smeared WZW-current Theorem 5.1. The internal TFPT code-token "MMST" and the legacy claim-ids `SEAM.MMST.*` / filenames `v*_mmst_*` denote *this* Osborne–Stottmeister theorem.)
- V. Morinelli, G. Morsella, A. Stottmeister, Y. Tanimoto, *Scaling limits of lattice quantum fields by wavelets*, Comm. Math. Phys. **387** (2021) 299–360, `arXiv:2010.11121`, `doi:10.1007/s00220-021-04152-5`; and *Operator-algebraic renormalization and wavelets*, Phys. Rev. Lett. **127** (2021) 230601. (The operator-algebraic (wavelet/Wilson–Kadanoff) renormalization *framework* the Osborne–Stottmeister theorem above builds on.)
- M. S. Adamo, Y. Moriwaki, Y. Tanimoto, *Osterwalder–Schrader axioms for unitary full vertex operator algebras*, `arXiv:2407.18222` (2024).
- M. S. Adamo, L. Giorgetti, Y. Tanimoto, *Rational and non-rational two-dimensional conformal field theories arising from lattices*, `arXiv:2506.01008` (2025); *Representations of conformal nets associated with infinite-dimensional groups*, `arXiv:2508.07109` (2025).
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